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MIL-M-28787/371
29 March 1991

MILITARY SPECIFICATION

MODULES, ANALOG, STANDARD ELECTRONIC

INTERFACE, DISCRETE MODULE, KEY CODE TAV

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic interface discrete module. This module is class I as specified in MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, Code SEA-55Z3, Department of the Navy, Washington, D.C. 20362-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification, standards and handbook. The following specification, standards and handbook form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

HANDBOOK

MILITARY

MIL-HDBK-217 - Reliability Prediction of Electronic Equipment.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a failure rate predicted in accordance with MIL-STD-1389. The predicted failure rate shall not exceed 2.796 failures per million operating hours when predicted in accordance with MIL-HDBK-217.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be TAV.

3.6.2 Size. The module size shall be 2A Format B.

3.6.3 Weight. The module shall have a maximum weight of 135 grams.

3.6.4 Contact pins. The module shall have 100 contact pins positioned 1 through 50 in one row and 51 through 100 in the other row.

3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

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TABLE I. Module electrical requirements.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		0 ± 5°C 4/		60 ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
A1	Isolation, initial	II	S	10	INF	10	INF	10	INF	MOhm
	end-of-life	II	Q	1	INF	1	INF	1	INF	MOhm
A2-A19	Isolation, initial	II	X	10	INF	10	INF	10	INF	MOhm
	end-of-life	II	Q	1	INF	1	INF	1	INF	MOhm
B1-B4	Continuity	III	X	0.0	1.0	0.0	1.0	0.0	1.0	Ohm
B5-B8	Continuity	III	Q	0.0	0.1	0.0	0.1	0.0	0.1	Ohm
C	Vcc stress	IV	Q	---------- Module shall pass all remaining tests ----------						
D1-D9	Ibd	V	S	-100	100	-100	100	-100	100	uA
D10,D11	Ibd	V	S	-200	200	-200	200	-200	200	uA
D12-D16	Ibd	V	S	-100	100	-100	100	-100	100	uA
E1-E16	Vic	VI	S	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	V
E17-E24	Vic	VI	X	-1.5	-0.5	-1.5	-0.5	-1.5	-0.5	V
F1-F8	Ios	VII	S	-130	-15	-130	-15	-130	-15	mA
F9	Ios	VII	S	-100	-15	-100	-15	-100	-15	mA
F10	Ios	VII	S	-130	-15	-130	-15	-130	-15	mA
G1-G14	Iih	VIII	X	0	20	0	20	0	20	uA
G15,G16	Iih	VIII	X	0	40	0	40	0	40	uA
H1-H8	Iil	IX	X	-400.0	-0.01	-400.0	-0.01	-400.0	-0.01	uA
H9	Iil	IX	X	-400.0	-30.0	-400.0	-30.0	-400.0	-30.0	uA
H10	Iil	IX	X	-400.0	-0.01	-400.0	-0.01	-400.0	-0.01	uA
H11-H14	Iil	IX	X	-410.0	-170.0	-410.0	-170.0	-410.0	-170.0	uA
H15,H16	Iil	IX	X	-800.0	-60.0	-800.0	-60.0	-800.0	-60.0	uA
J1-J32	Voh	X	X	2.4	4.5	2.4	4.5	2.4	4.5	V
J33,J34	Voh	X	X	2.5	4.5	2.5	4.5	2.5	4.5	V
K1-K35	Vol	XI	X	0.0	0.4	0.0	0.4	0.0	0.4	V
L1-L8	Iozh	XII	X	-20	20	-20	20	-20	20	uA

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		0 ± 5°C 4/		60 ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
M1-M8	Iozl	XIII	X	-20	20	-20	20	-20	20	uA
N1	Iccl	XIV	X	200	380	200	380	200	380	mA
N2	Icch	XIV	X	200	380	200	380	200	380	mA
P1,P2	Functional	XV	X	See table XV						
Q1,Q2	Dynamic	XVI	S,X	See table XVI						
R	PSFC	XVIII	S	1.59	2.69	1.59	2.69	1.59	2.69	uF
S	PSTR	XIX	Q	0	400	0	400	0	400	mV
T1-T58	C1/o	XX	Q	2	100	2	100	2	100	pF
U1-U16	Switch "ON" resistance	XXI	X	0	145	0	185	0	185	mV
U17-U32	Switch "ON" resistance	XXI		-145	0	-185	0	-185	0	mV
V1-V18	FET switch leakage current	XXII	X	0	25	0	110	0	110	uA
W1-W16	FET switch power "OFF" leakage current	XXIII	X	0	25	0	110	0	110	uA
X1-X8	Input sense diode, forward voltage	XXIV	X	0.0	1.2	0.0	1.2	0.0	1.2	V
Y1-Y8	Input sense diode, reverse leakage current	XXV	X	-10	0	-10	0	-10	0	uA

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		0 ± 5°C 4/		60 ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
Z1-Z16	Zener voltage	XXVI	X	73.8	90.2	73.8	90.2	73.8	90.2	V
AA1	Constant current diode	XXVII	X	0.675	0.825	0 510	0.956	0.510	0.956	mA
AA2	Constant current diode	XXVII	X	3.51	4.60	2.33	5.304	2.33	5.304	mA
BB1-BB8	Disable time, tPLZ	XXVIII	Q	7	69	7	69	7	69	nS
BB9-BB16	Disable time, tPHZ	XXVIII	Q	7	65	7	65	7	65	nS
CC1-CC16	Input sense current	XXIX	X	6.5	7.5	6.5	7.5	6.5	7.5	mA
CC17-CC32	Input sense current	XXIX	X	2.8	3.5	2.8	3.5	2.8	3.5	mA
CC33-CC40	Input sense current	XXIX	X	11.0	14.0	11.0	14.0	11.0	14.0	mA
CC41-CC48	Input sense current	XXIX	X	15.3	17.0	15.3	17.0	15.3	17.0	mA
CC49-CC56	Input sense current	XXIX	X	1.5	3.5	1.5	3.5	1.5	3.5	mA
CC57-CC64	Input sense current	XXIX	X	3.5	5.0	3.5	5.0	3.5	5.0	mA

- See footnotes on next page.

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1/ "X" parameters shall be 100 percent tested during the module acceptance test.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified in the test table, limits are for initial and end-of-life.

5/ Abbreviations used in table I are defined in 6.1.

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3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/371-1. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The functional block diagram shall conform to figure 1. The pin assignments shall conform to figure 2.

3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. The module shall be capable of conforming to the requirements in table I for a power supply voltage (Vcc) of 5.0 ± 0.5 V dc, except the output short circuit current limits apply only for a power supply voltage of 5.5 V dc ± 1 percent.

3.11 Parallel outputs. Paralleling of the TTL logic outputs is not permissible. However, parallel connection of RAX with RBX is permissible for dc currents where (X = 0 to 7).

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified herein, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.

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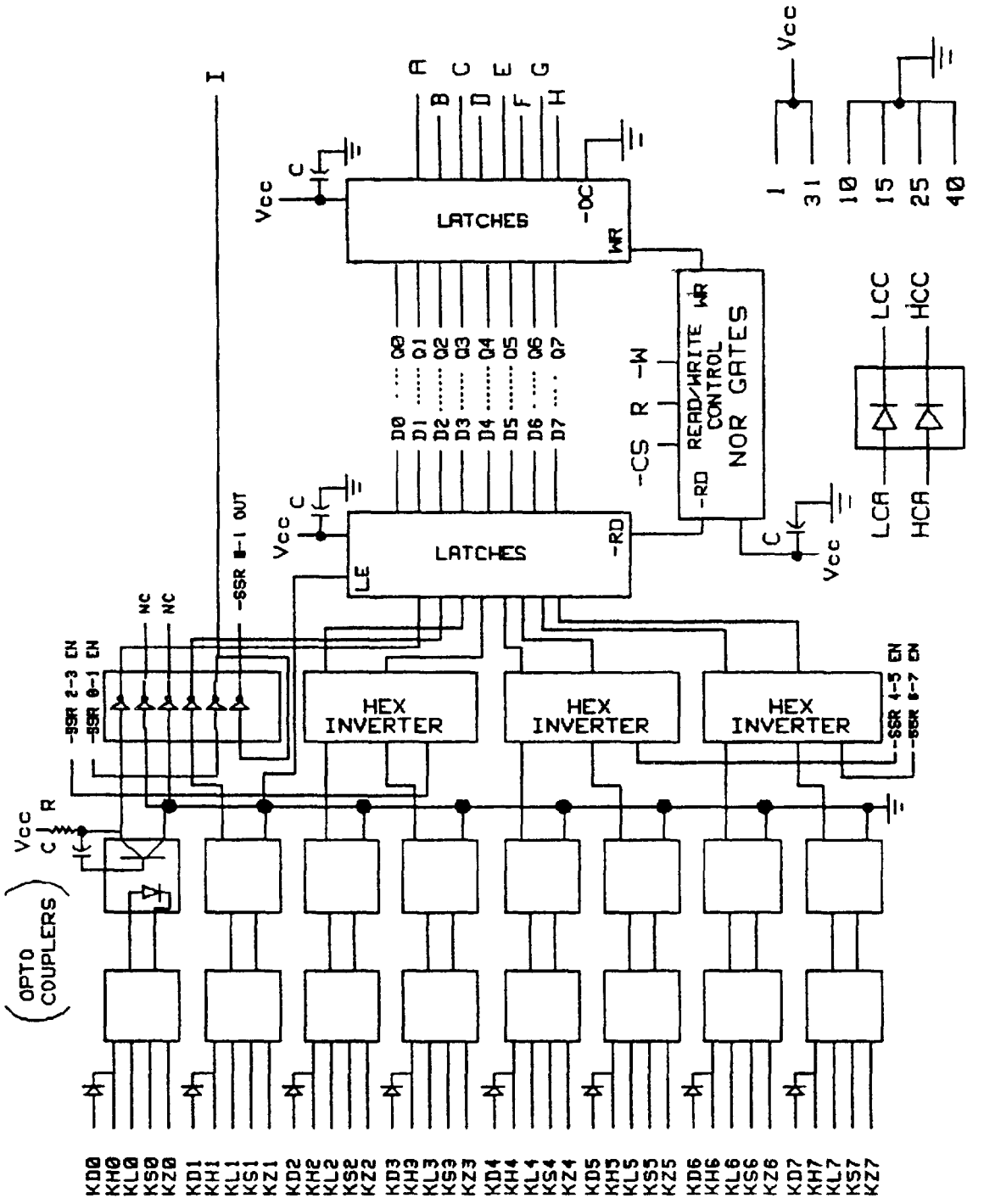


FIGURE 1. TAV functional block diagram.

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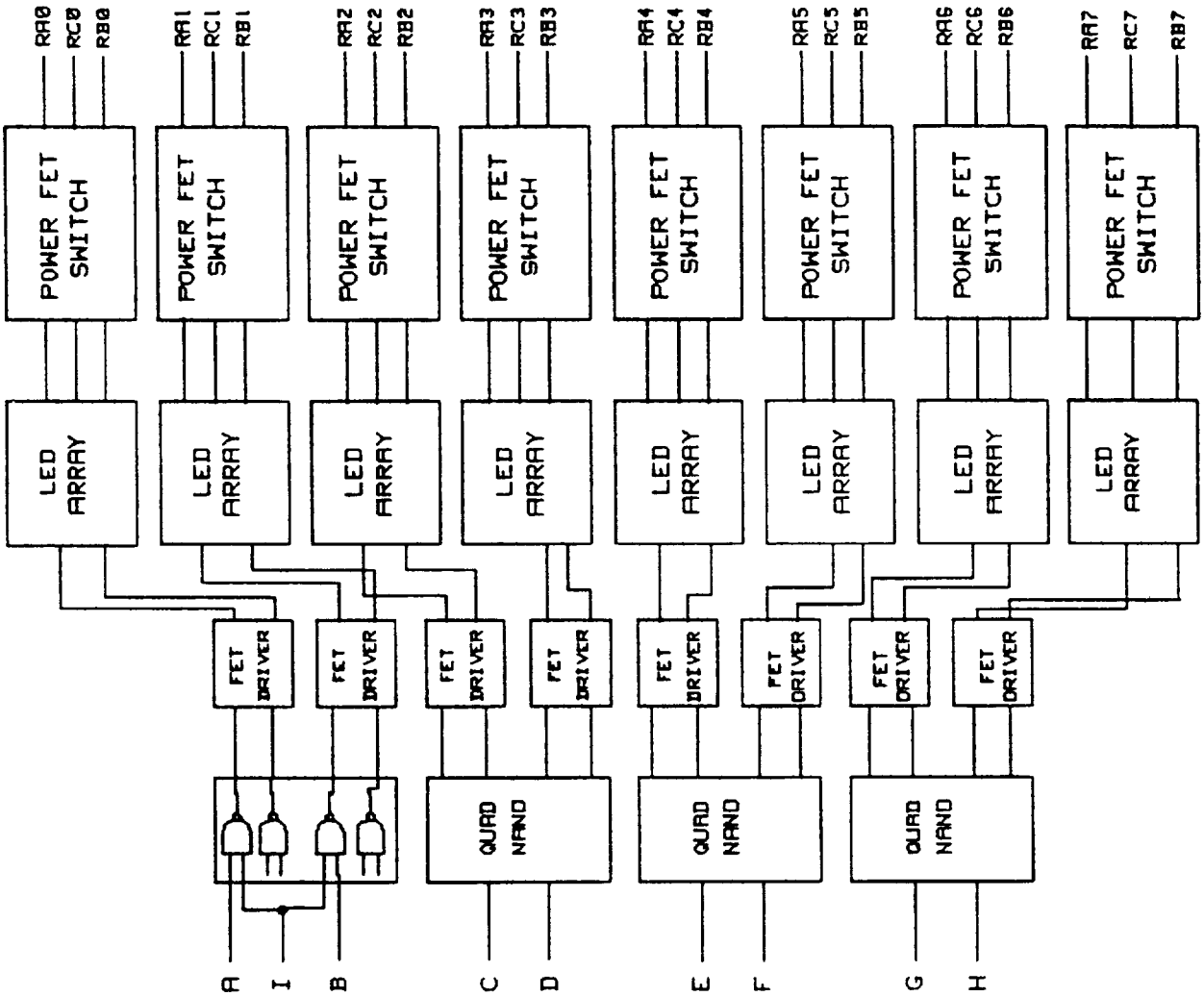


FIGURE 1. TAV functional block diagram - Continued.

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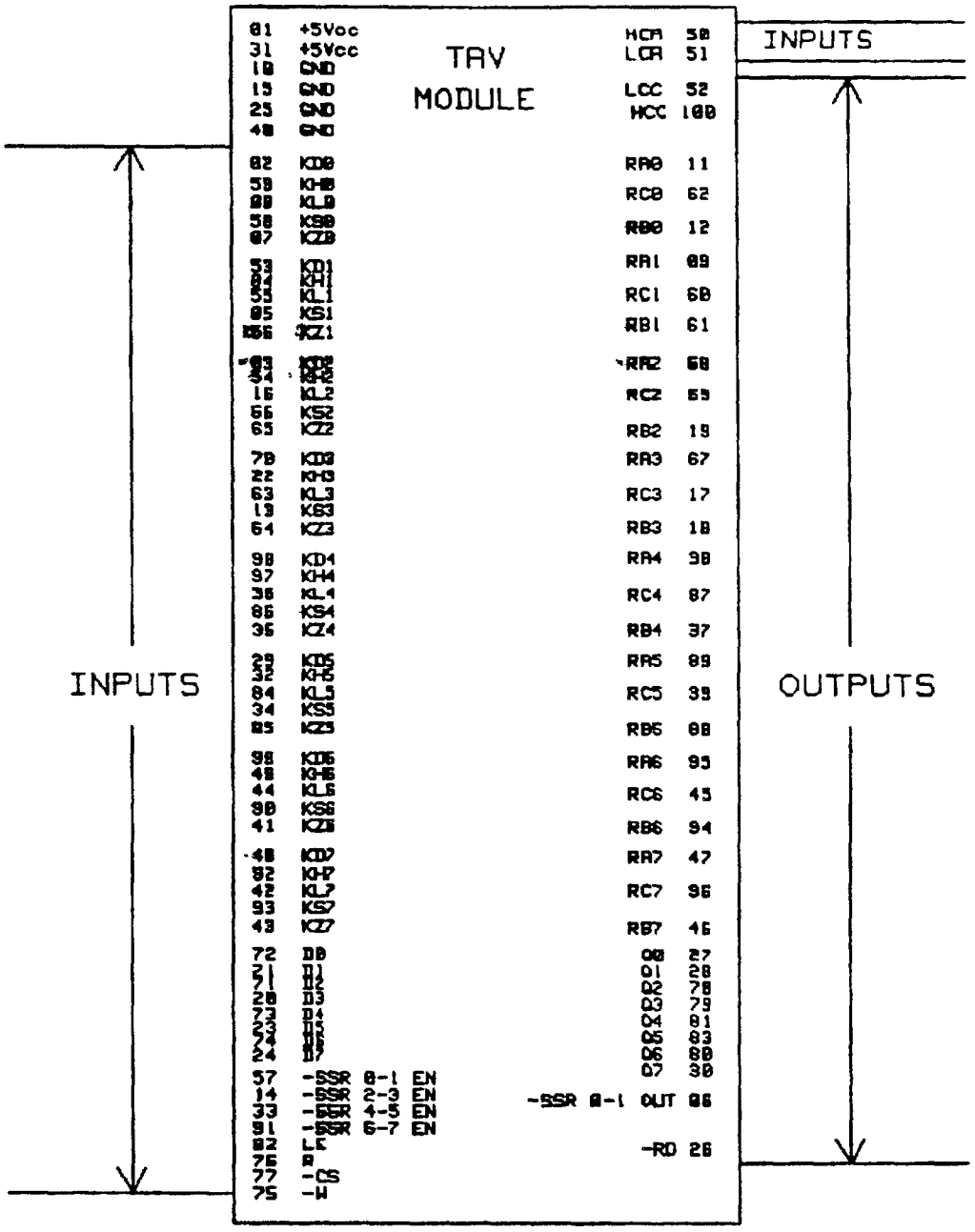


FIGURE 2. TAV pin assignment.

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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through CC shall be performed in any order with the following exceptions. Test conditions A, C, D, E and F are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, C, D, E or F are part of a sequence of tests, the order of testing shall be as follows: A, C, D, E and F where applicable, followed by the remaining tests in the sequence.

4.4.3 Multiple testing of pins. Singular pins listed multiple times in a test table indicate multiple devices connected to the pins. The module shall be tested so that each device connected to a pin is exercised and tested.

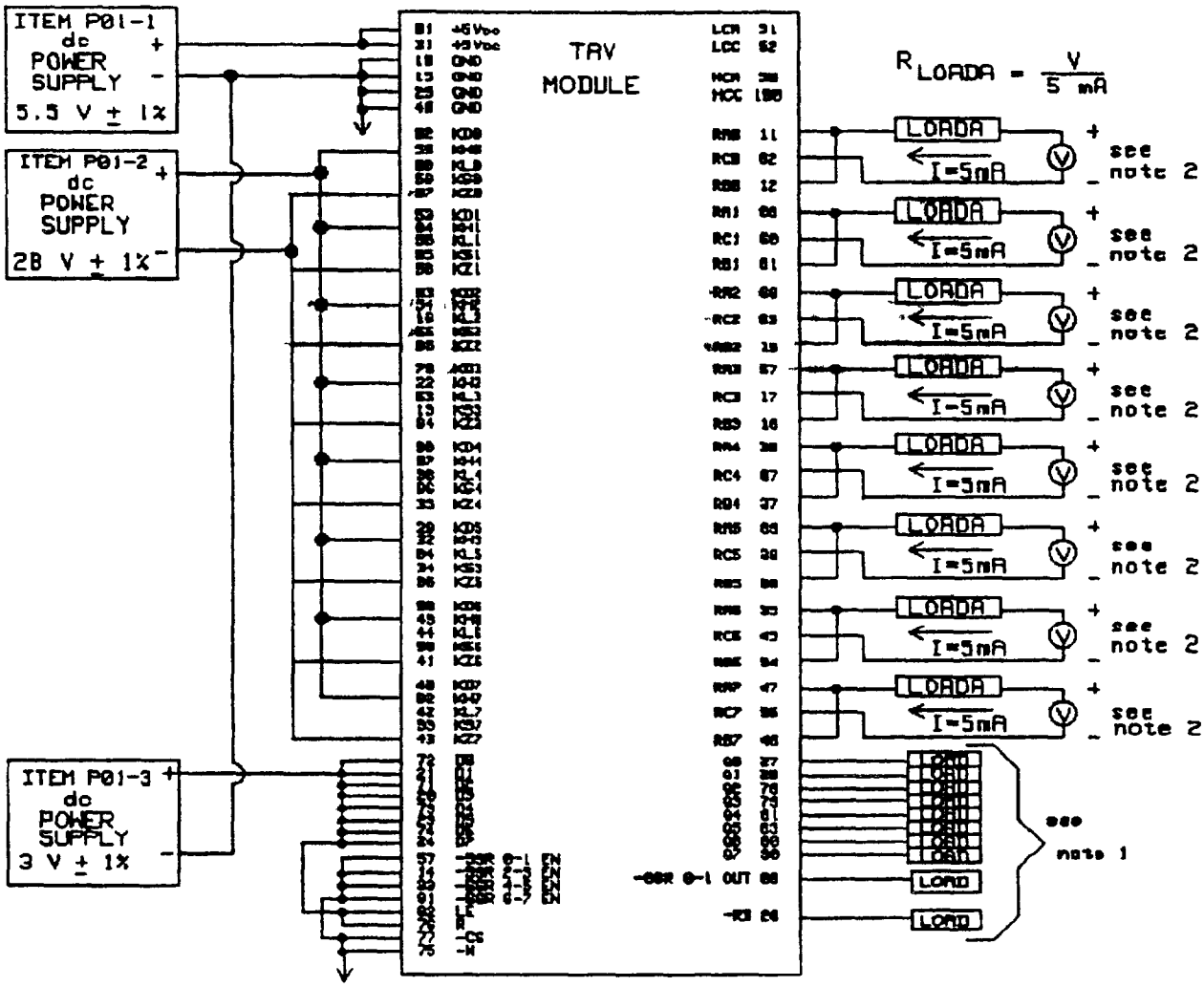
4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through CC in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B, G through CC (except T) shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated as shown on figure 3.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 4.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 5. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition R, shall be performed using the initial limits at 25 ± 5°C ambient.

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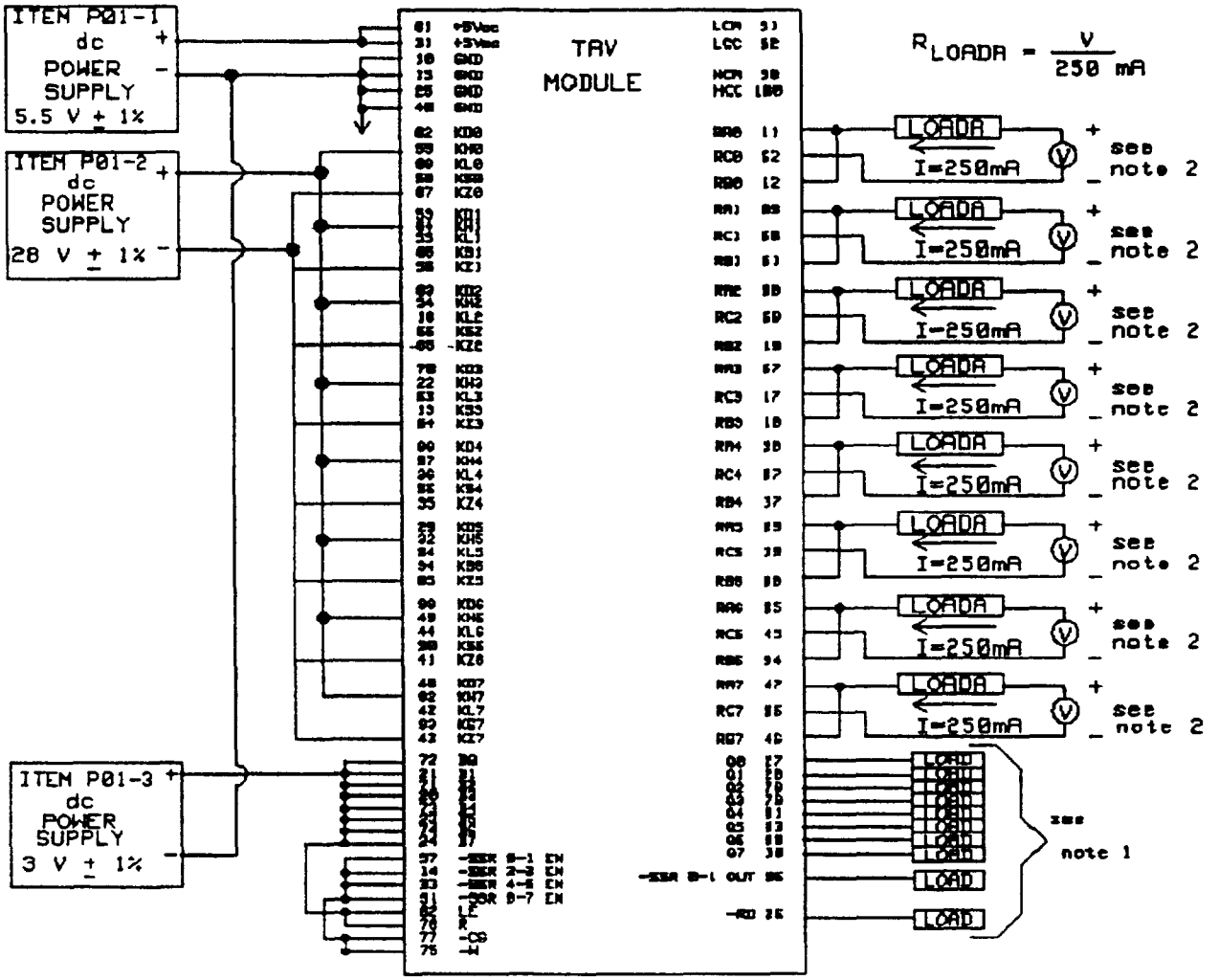


NOTES:

1. The load for each output is specified on figure 6.
2. 2.5 V ± 1 percent.

FIGURE 3. TAV operating temperature and life test circuit.

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NOTES:

1. The load for each output is specified on figure 6.
2. 2.5 V ± 1 percent.

FIGURE 4. TAV component temperature test circuit.

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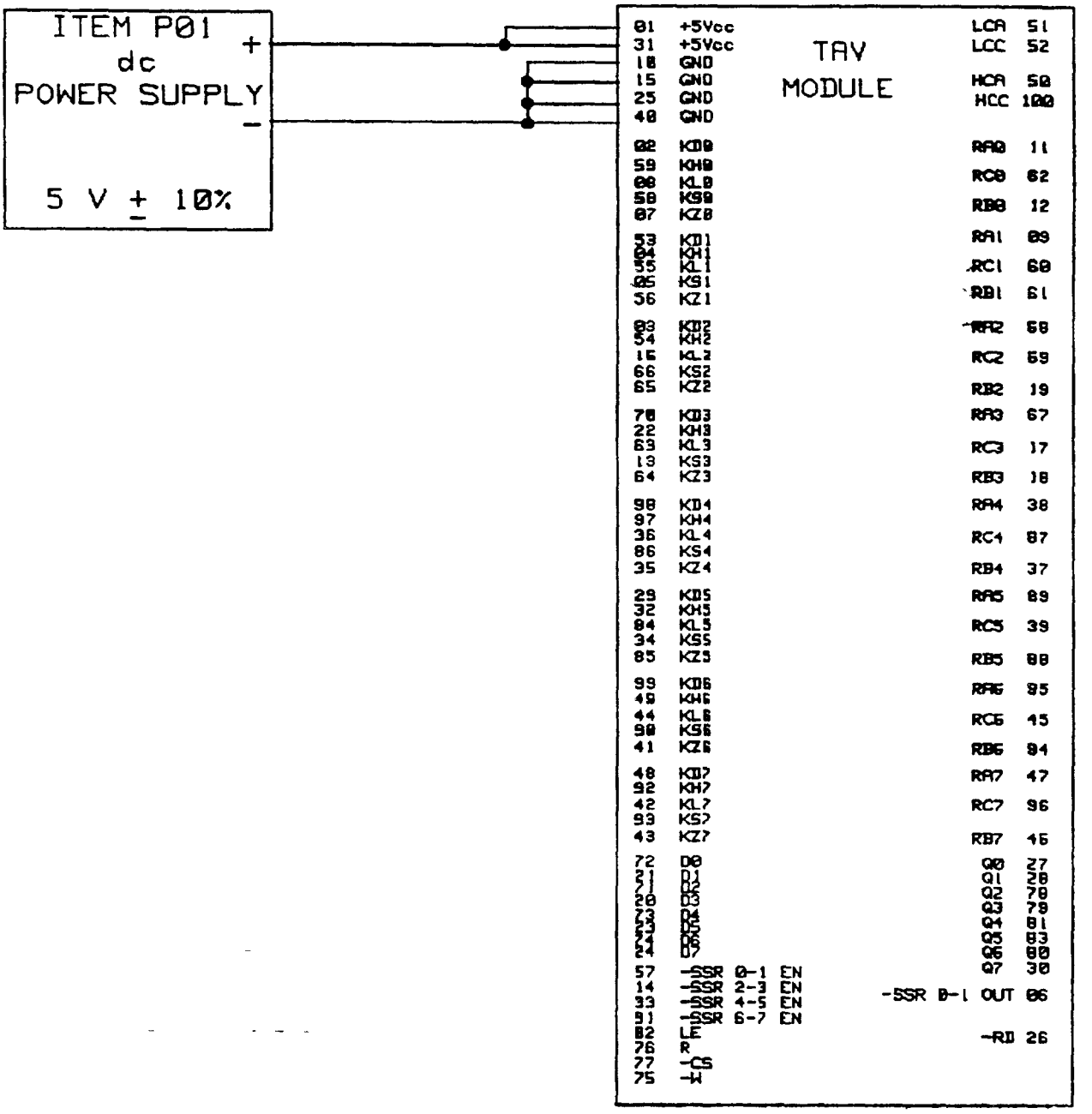


FIGURE 5. TAV vibration test circuit.

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4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3. Following the life test and while still at the maximum class temperature, electrical tests B, G through CC (except T) shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests B, G through CC (except T) shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

4.5.4 Thermal shock test. The thermal shock test shall be conducted in accordance with MIL-M-28787. The module shall be unpowered during this test. Following the thermal shock test, the electrical tests in table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25 ± 5°C ambient.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Negative (-) current is current directed out of the module pin and positive (+) current is current directed into the module pin. Abbreviations used in this specification are defined as follows:

(a) Cin = input capacitance.
(b) Ci/o = input and output capacitance.
(c) Cout = output capacitance.
(d) FET = field effect transistor.
(e) Ibd = input breakdown current.
(f) Icc = power supply current.
(g) Icch = power supply current high level.
(h) Iccl = power supply current low level.
(i) Iih = input high level current.
(j) Iil = input low level current.
(k) Iin = input forcing current.
(l) Ioh = output high level current.
(m) Iol = output low level current.
(n) Ios = output short circuit current.
(o) Iozh = output high level leakage current.
(p) Iozl = output low level leakage current.
(q) NA = not applicable.
(r) PSFC = power supply filter capacitance.
(s) PSTA = power supply transient amplitude.
(t) Vic = input clamp diode voltage.
(u) Vih = input high level voltage.
(v) Vil = input low level voltage.
(w) Vin = input forcing voltage.
(x) Voh = output high level voltage.
(y) Vol = output low level voltage.
(z) Vout = output forcing voltage.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Module description. The following electrical parameters reflect the 0°C to 60°C end-of-life parameters in table I. Operating power supply voltage range: 4.5 V dc to 5.5 V dc. Module power dissipation: 2.09 W (max); 1.495 W (typical).

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6.3.1 Input parameters.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (uA)	Cin max (pF)	Iil max (mA)	Iih min (mA)
D0	-1.5	0.7	2.0	-0.4	0.02	100	100	NA	NA
D1	-1.5	0.7	2.0	-0.4	0.02	100	100	NA	NA
D2	-1.5	0.7	2.0	-0.4	0.02	100	100	NA	NA
D3	-1.5	0.7	2.0	-0.4	0.02	100	100	NA	NA
D4	-1.5	0.7	2.0	-0.4	0.02	100	100	NA	NA
D5	-1.5	0.7	2.0	-0.4	0.02	100	100	NA	NA
D6	-1.5	0.7	2.0	-0.4	0.02	100	100	NA	NA
D7	-1.5	0.7	2.0	-0.4	0.02	100	100	NA	NA
LE	-1.5	0.7	2.0	-0.4	0.02	100	100	NA	NA
-W	-1.5	0.7	2.0	-0.8	0.04	200	100	NA	NA
-CS	-1.5	0.7	2.0	-0.8	0.04	200	100	NA	NA
R	-1.5	0.7	2.0	-0.4	0.02	100	100	NA	NA
-SSR 0 EN	-1.5	0.5	2.0	-0.36	0.02	100	100	NA	NA

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (uA)	Cin max (pF)	Iil max (mA)	Iih min (mA)
-SSR 1 EN	-1.5	0.5	2.0	-0.36	0.02	100	100	NA	NA
-SSR 2 EN	-1.5	0.5	2.0	-0.36	0.02	100	100	NA	NA
-SSR 3 EN	-1.5	0.5	2.0	-0.36	0.02	100	100	NA	NA
-SSR 4 EN	-1.5	0.5	2.0	-0.36	0.02	100	100	NA	NA
-SSR 5 EN	-1.5	0.5	2.0	-0.36	0.02	100	100	NA	NA
-SSR 6 EN	-1.5	0.5	2.0	-0.36	0.02	100	100	NA	NA
-SSR 7 EN	-1.5	0.5	2.0	-0.36	0.02	100	100	NA	NA
KS0	1.5	NA	NA	-240	240	NA	100	0.9	3.0
KS1	1.5	NA	NA	-240	240	NA	100	0.9	3.0
KS2	1.5	NA	NA	-240	240	NA	100	0.9	3.0
KS3	1.5	NA	NA	-240	240	NA	100	0.9	3.0
KS4	1.5	NA	NA	-240	240	NA	100	0.9	3.0
KS5	1.5	NA	NA	-240	240	NA	100	0.9	3.0

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (uA)	Cin max (pF)	Iil max (mA)	Iih min (mA)
KS6	1.5	NA	NA	-240	240	NA	100	0.9	3.0
KS7	1.5	NA	NA	-240	240	NA	100	0.9	3.0
KL0	NA	0.9	2.4	NA	NA	NA	100	NA	NA
KL1	NA	0.9	2.4	NA	NA	NA	100	NA	NA
KL2	NA	0.9	2.4	NA	NA	NA	100	NA	NA
KL3	NA	0.9	2.4	NA	NA	NA	100	NA	NA
KL4	NA	0.9	2.4	NA	NA	NA	100	NA	NA
KL5	NA	0.9	2.4	NA	NA	NA	100	NA	NA
KL6	NA	0.9	2.4	NA	NA	NA	100	NA	NA
KL7	NA	0.9	2.4	NA	NA	NA	100	NA	NA
KH0	NA	4.0	12.0	NA	NA	NA	100	NA	NA
KH1	NA	4.0	12.0	NA	NA	NA	100	NA	NA
KH2	NA	4.0	12.0	NA	NA	NA	100	NA	NA

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (uA)	Cin max (pF)	Iil max (mA)	Iih min (mA)
KH3	NA	4.0	12.0	NA	NA	NA	100	NA	NA
KH4	NA	4.0	12.0	NA	NA	NA	100	NA	NA
KH5	NA	4.0	12.0	NA	NA	NA	100	NA	NA
KH6	NA	4.0	12.0	NA	NA	NA	100	NA	NA
KH7	NA	4.0	12.0	NA	NA	NA	100	NA	NA
KD0	NA	4.0	12.0	NA	NA	NA	100	NA	NA
KD1	NA	4.0	12.0	NA	NA	NA	100	NA	NA
KD2	NA	4.0	12.0	NA	NA	NA	100	NA	NA
KD3	NA	4.0	12.0	NA	NA	NA	100	NA	NA
KD4	NA	4.0	12.0	NA	NA	NA	100	NA	NA
KD5	NA	4.0	12.0	NA	NA	NA	100	NA	NA
KD6	NA	4.0	12.0	NA	NA	NA	100	NA	NA
KD7	NA	4.0	12.0	NA	NA	NA	100	NA	NA

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6.3.2 Output parameters.

Signal	Voh min (V)	Vol max (V)	Ioh max (uA)	Iol max (mA)	Iozh max (uA)	Iozl min (uA)	Ios min (mA)	Cout max (pF)
Q0	2.5	0.4	-1000	12.0	20	-20	-130	100
Q1	2.5	0.4	-1000	12.0	20	-20	-130	100
Q2	2.5	0.4	-1000	12.0	20	-20	-130	100
Q3	2.5	0.4	-1000	12.0	20	-20	-130	100
Q4	2.5	0.4	-1000	12.0	20	-20	-130	100
Q5	2.5	0.4	-1000	12.0	20	-20	-130	100
Q6	2.5	0.4	-1000	12.0	20	-20	-130	100
Q7	2.5	0.4	-1000	12.0	20	-20	-130	100
-SSR 0-1 OUT	2.5	0.4	400	4.0	NA	NA	-100	100
-RD	2.5	0.4	-380	3.6	NA	NA	-130	100

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6.3.3 Dynamic parameters. The dynamic parameters appear with the dynamic test.

6.4 Module function. The interface discrete module is a solid state switch and sense module having two levels of electro-optic isolation and data bus control. Each interface Input/Output (I/O) operates at various voltages/currents of ac or dc polarity. The module consists of sixteen discrete interface channels with eight discrete inputs and eight discrete outputs that can be connected to the microprocessor bus interface. Each discrete input signal is optically isolated from the bus interface by an opto-coupler. Each of the inputs can be operated in only one of four input modes: current sense, low voltage sense, high voltage sense, and diode input high voltage sense. Each discrete output is made up of a Solid State Relay (SSR). The SSR is activated by a bus interface signal that controls Metal-Oxide-Semiconductor FET (MOSFET) drive circuitry for a photo-voltaic diode array which provides an isolated gate voltage through the output MOSFET switches. The bus interface consists of two octal tri-state transparent latches for reading the status of each discrete input and controlling each discrete output. This enables the user to either connect the two discrete I/Os together on a data bus or have independent control of the discrete I/O. The latches can be enabled so that they pass data without the need for read and write strobes. This feature allows the module to be used without a supporting microprocessor. In addition, each pair of SSR contacts can be controlled from an enable input.

6.4.1 Functional description of I/O pins.

Signal name	Pin number	I/O	Functional description
KD0	2	I	High voltage sense diode inputs.
KD1	53	I	
KD2	3	I	
KD3	70	I	
KD4	98	I	
KD5	29	I	
KD6	99	I	
KD7	48	I	
KH0	59	I	High voltage sense inputs.
KH1	4	I	
KH2	54	I	

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6.4.1 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
KH3	22	I	High voltage sense inputs.
KH4	97	I	
KH5	32	I	
KH6	49	I	
KH7	92	I	
KL0	8	I	Low voltage sense inputs.
KL1	55	I	
KL2	16	I	
KL3	63	I	
KL4	36	I	
KL5	84	I	
KL6	44	I	
KL7	42	I	
KS0	58	I	Current sense inputs.
KS1	5	I	
KS2	66	I	
KS3	13	I	
KS4	86	I	
KS5	34	I	
KS6	90	I	
KS7	93	I	
RA0	11	O	Solid state relay "A" outputs.
RA1	9	O	
RA2	68	O	
RA3	67	O	
RA4	38	O	
RA5	89	O	
RA6	95	O	
RA7	47	O	
RB0	12	O	Solid state relay "B" outputs.
RB1	61	O	
RB2	19	O	
RB3	18	O	
RB4	37	O	
RB5	88	O	
RB6	94	O	
RB7	46	O	

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6.4.1 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
RC0	62	O	Solid state relay "common" outputs.
RC1	60	O	
RC2	69	O	
RC3	17	O	
RC4	87	O	
RC5	39	O	
RC6	45	O	
RC7	96	O	
D0	72	I	Data bus switch control inputs - D0 is the least significant bit, and D7 is the most significant bit.
D1	21	I	
D2	71	I	
D3	20	I	
D4	73	I	
D5	23	I	
D6	74	I	
D7	24	I	
Q0	27	O	Data bus sense circuit output - Q0 is the least significant bit, and Q7 is the most significant bit.
Q1	28	O	
Q2	78	O	
Q3	79	O	
Q4	81	O	
Q5	83	O	
Q6	80	O	
Q7	30	O	
-SSR 0-1 OUT	6	O	Enable output signal.
-SSR 0-1 EN	57	I	Active low SSR enable for channels 0 and 1.
-SSR 2-3 EN	14	I	Active low SSR enable for channels 2 and 3.
-SSR 4-5 EN	33	I	Active low SSR enable for channels 4 and 5.
-SSR 6-7 EN	91	I	Active low SSR enable for channels 6 and 7.

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6.4.1 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
-RD	26	O	Active low read command - output to data bus.
-W	75	I	Active low write command - input from data bus.
R	76	I	Active high read command - input from data bus.
-CS	77	I	Active low chip select command - input from data bus.
LE	82	I	Active high latch enable command - input from data bus.
HCA	50	I	High current regulator diode - anode.
HCC	100	O	High current regulator diode - cathode.
LCA	51	I	Low current regulator diode - anode.
LCC	52	O	Low current regulator diode - cathode.
KZ0	7	I	Sense circuit, grounds.
KZ1	56	I	
KZ2	65	I	
KZ3	64	I	
KZ4	35	I	
KZ5	85	I	
KZ6	41	I	
KZ7	43	I	

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6.5 Description of tests. The following tables and associated figures describe the tests to be performed.

(a) Isolation test - table II.

(b) Continuity test - table III.

(c) Vcc stress test - table IV.

(d) Ibd test - table V.

(e) Vic test - table VI.

(f) Ios test - table VII.

(g) Iih test - table VIII.

(h) Iil test - table IX.

(i) Voh test - table X, figure 6.

(j) Vol test - table XI, figure 6.

(k) Iozh test - table XII.

(l) Iozl test - table XIII.

(m) Power supply current (Iccl and Icch) test - table XIV.

(n) Functional test - table XV, figure 7.

(o) Dynamic test - tables XVI and XVII, figure 8.

(p) PSFC test - table XVIII.

(q) PSTA test - table XIX, figure 9.

(r) Input/Output capacitance test - table XX.

(s) Switch "ON" resistance test - table XXI, figures 10 and 11.

(t) FET switch leakage current test - table XXII, figure 12.

(u) FET switch power "OFF" leakage current test - table XXIII, figure 13.

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(v) Input sense diode, forward voltage test - table XXIV, figure 14.

(w) Input sense diode, reverse leakage current test - table XXV, figure 15.

(x) Zener voltage test - table XXVI; figure 16.

(y) Constant current diode test - table XXVII, figure 17.

(z) Disable time (tPLZ and tPHZ) test - table XXVIII, figures 18 and 19.

(aa) Input sense current test - table XXIX; figure 20.

6.6 Functional test vectors. Functional test vectors are not printed herein because of their extreme length; and, even when presented as a hard copy, they require conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be generated in a form compatible with test system architecture (magnetic tape format usually preferred). For reference purposes, the Naval Weapons Support Center (Code 607) maintains a limited quantity of dated, hard copy printouts of the functional test vectors and these may be obtained upon request.

Custodians:
Army - ER
Navy - SH
Air Force - 85

Review activities:
Army - AT
Navy - AS, MC
Air Force - 13, 19
DLA - ES

Preparing activity:
Navy - SH

(Project 5963-0064)

Agent: NW

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TABLE II. Isolation test.

Test cond	Test description	Connect		25 ± 5°C Initial limits		Units	Notes
		A to pin	B to pin	Min	Max		
A1	Isolation, frame to all other pins	All pins	Frame	10	INF	MOhms	1-7
A2	Circuit 1 isolation, input sense circuit network	2,7,8,58,59	All other remaining pins (tied together)				
A3	Circuit 2 isolation, input sense circuit network	4,5,53,55, 56					
A4	Circuit 3 isolation, input sense circuit network	3,16,54,65, 66					
A5	Circuit 4 isolation, input sense circuit network	13,22,63, 64,70					
A6	Circuit 5 isolation, input sense circuit network	35,36,96, 97,98					
A7	Circuit 6 isolation, input sense circuit network	29,32,34, 84,85					
A8	Circuit 7 isolation, input sense circuit network	41,44,49, 90,99	All other remaining pins (tied together)	10	INF	MOhms	1-7

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TABLE II. Isolation test - Continued.

Test cond	Test description	Connect		25 + 5°C Initial limits		Units	Notes
		A to pin	B to pin	Min	Max		
A9	Circuit 8 isolation, input sense circuit network	42,43,48, 92,93	All other remaining pins (tied together)	10	INF	MOhms	1-7
A10	Circuit 9 isolation, output relay circuit network	11,12,62					
A11	Circuit 10 isolation, output relay circuit network	9,60,61					
A12	Circuit 11 isolation, output relay circuit network	19,68,69					
A13	Circuit 12 isolation, output relay circuit network	17,18,67					
A14	Circuit 13 isolation, output relay circuit network	37,38,87					
A15	Circuit 14 isolation, output relay circuit network	39,88,89					
A16	Circuit 15 isolation, output relay circuit network	45,94,95	All other remaining pins (tied together)	10	INF	MOhms	1-7

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TABLE II. Isolation test - Continued.

Test cond	Test description	Connect		25 ± 5°C Initial limits		Units	Notes
		A to pin	B to pin	Min	Max		
A17	Circuit 16 isolation, output relay circuit network	46,47,96	All other remaining pins (tied together)	10	INF	MOhms	1-7
A18	Isolation, low constant current diode	51,52					
A19	Isolation, high constant current diode	50,100	All other remaining pins (tied together)	10	INF	MOhms	1-7

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NOTES:

1. Isolation test tolerance table.

Condition	Value	Accuracy	Test condition
Megohm bridge voltage	100 V	± 10 percent	A1-A19
Isolation measurement (see note 6)	NA NA	± 110 KOhm ± 11 KOhm	A1-A19 A1-A19

2. The ohmmeter used shall be item R06.

3. In the setup conditions in table II, 'A' shall correspond to the positive terminal of the megohm bridge, and 'B' shall correspond to the negative terminal.

4. Warning: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

5. When testing for frame isolation, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of the megohm bridge.

6. The minimum end-of-life limit is 1 MOhm.

7. Abbreviations used are defined in 6.1.

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TABLE III. Continuity test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
B1	Continuity	1	31	0.0	1.0	Ohm	1-4
B2	Continuity	10	15				
B3	Continuity	15	25				
B4	Continuity	25	40	0.0	1.0	Ohm	1-4
B5	Continuity	1	31	0.0	0.1	Ohm	1-4
B6	Continuity	10	15				
B7	Continuity	15	25				
B8	Continuity	25	40	0.0	0.1	Ohm	1-4

NOTES:

1. Continuity test tolerance table.

Condition	Value	Accuracy	Test condition
Continuity measurement	NA NA	± 11 mOhm ± 1.1 mOhm	B1-B4 B5-B8

2. The ohmmeter used shall be item R07.

3. In the setup conditions in table III, 'A' shall correspond to the positive terminal of the ohmmeter, and 'B' shall correspond to the negative terminal.

4. Abbreviations used are defined in 6.1.

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TABLE IV. Power supply stress test.

Test cond	Power supply	Vin (V)	Connect Positive lead to	Connect Negative lead to	25 ± 5°C Initial limits	Notes
C	Vcc	7.0	1,31	10,15 25,40	Module shall pass all remaining tests.	1-5

NOTES:

1. Power supply stress test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	7.0 V	+0 mV,-70 mV	C
Power supply current limit	0.684 A	NA	C

2. There shall be no power supply voltage overshoot on this test.

3. Connect the power supplies to the module as indicated in table IV.

4. When performing the power supply stress test, no measurements are taken. This condition shall be maintained for a minimum of 5 seconds.

5. Abbreviations used are defined in 6.1.

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TABLE V. Input breakdown current test.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D1	D0	72	5.5	-100	100	uA	1-3
D2	D1	21					
D3	D2	71					
D4	D3	20					
D5	D4	73					
D6	D5	23					
D7	D6	74					
D8	D7	24					
D9	LE	82	5.5	-100	100	uA	1-3
D10	-W	75	5.5	-200	200	uA	1-3
D11	-CS	77	5.5	-200	200	uA	1-3
D12	R	76	5.5	-100	100	uA	1-3
D13	-SSR 0-1 EN	57					
D14	-SSR 2-3 EN	14					
D15	-SSR 4-5 EN	33					
D16	-SSR 6-7 EN	91	5.5	-100	100	uA	1-3

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NOTES:

1. Input breakdown current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	D1-D16
Power supply current limit	0.57 A	NA	D1-D16
Forcing voltage, (Vin)	5.5 V	± 55 mV	D1-D16
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 D1-D16 D1-D16
Input current measurement	 NA NA NA	 ± 2.2 uA ± 2.2 uA ± 2.2 uA	 D1-D9 D10,D11 D12-D16

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table V and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VI. Input clamp diode voltage test.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
E1	D0	72	-18.0	-1.5	-0.4	V	1-5
E2	D1	21					
E3	D2	71					
E4	D3	20					
E5	D4	73					
E6	D5	23					
E7	D6	74					
E8	D7	24					
E9	LE	82					
E10	-W	75					
E11	-CS	77					
E12	R	76					
E13	-SSR 0-1 EN	57					
E14	-SSR 2-3 EN	14					
E15	-SSR 4-5 EN	33					
E16	-SSR 6-7 EN	91	-18.0	-1.5	-0.4	V	1-5
E17	KS0	58	-200.0	-1.5	-0.5	V	1-5
E18	KS1	5					
E19	KS2	66					
E20	KS3	13	-200.0	-1.5	-0.5	V	1-5

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TABLE VI. Input clamp diode voltage test - Continued.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
E21	KS4	86	-200.0	-1.5	-0.5	V	1-5
E22	KS5	34					
E23	KS6	90					
E24	KS7	93	-200.0	-1.5	-0.5	V	1-5

NOTES:

1. Input clamp diode voltage test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	E1-E24
Power supply current limit	0.57 A	NA	E1-E24
Forcing current, (Iin)	-18 mA -200 mA	± 1.8 mA ± 20.0 mA	E1-E16 E17-E24
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	E1-E24 E1-E24
Input voltage measurement	NA NA	± 12.1 mV ± 8.8 mV	E1-E16 E17-E24

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table VI and the voltage then measured.

3. Vin shall not exceed 1.5 V for test conditions E1 through E16 and 2.0 V for test conditions E17 through E24.

4. For test conditions E17 through E24 measure KSx with respect to KZx (x=0 to 7).

5. Abbreviations used are defined in 6.1.

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TABLE VII. Output short circuit current test.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F1	Q0	27	0.0	-130	-15	mA	1-4
F2	Q1	28					
F3	Q2	78					
F4	Q3	79					
F5	Q4	81					
F6	Q5	83					
F7	Q6	80					
F8	Q7	30	0.0	-130	-15	mA	1-4
F9	-SSR 0-1 OUT	6	0.0	-100	-15	mA	1-4
F10	-RD	26	0.0	-130	-15	mA	1-4

NOTES:

1. Output short circuit current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	F1-F10
Power supply current limit	0.57 A	NA	F1-F10
Forcing voltage, (Vout)	0.0 V	± 20 mV	F1-F10

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 F1-F10 F1-F10
Output current measurement	 NA NA NA	 ± 1.27 mA ± 0.94 mA ± 1.27 mA	 F1-F8 F9 F10

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VII and the current then measured. The forcing voltage undershoot shall not exceed 100 mV.

3. When performing this test, only one output shall be tested at a time, and the duration of momentarily grounding each output shall not exceed 1 second.

4. Abbreviations used are defined in 6.1.

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TABLE VIII. Input current, high level, test.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
G1	D0	72	2.7	0	20	uA	1-3
G2	D1	21					
G3	D2	71					
G4	D3	20					
G5	D4	73					
G6	D5	23					
G7	D6	74					
G8	D7	24					
G9	R	76					
G10	LE	82					
G11	-SSR 0-1 EN	57					
G12	-SSR 2-3 EN	14					
G13	-SSR 4-5 EN	33					
G14	-SSR 6-7 EN	91	2.7	0	20	uA	1-3
G15	-W	75	2.7	0	40	uA	1-3
G16	-CS	77	2.7	0	40	uA	1-3

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NOTES:

1. Input current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	G1-G16
Power supply current limit	0.57 A	NA	G1-G16
Forcing voltage, (Vih)	2.7 V	± 27 mV	G1-G16
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	G1-G16 G1-G16
Input current measurement	NA NA	± 0.22 uA ± 0.44 uA	G1-G14 G15,G16

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE IX. Input current, low level, test.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
H1	D0	72	0.4	-400.0	-0.01	uA	1-3
H2	D1	21					
H3	D2	71					
H4	D3	20					
H5	D4	73					
H6	D5	23					
H7	D6	74					
H8	D7	24	0.4	-400.0	-0.01	uA	1-3
H9	R	76	0.4	-400.0	-30.0	uA	1-3
H10	LE	82	0.4	-400.0	-0.01	uA	1-3
H11	-SSR 0-1 EN	57	0.4	-410.0	-170.0	uA	1-3
H12	-SSR 2-3 EN	14					
H13	-SSR 4-5 EN	33					
H14	-SSR 6-7 EN	91	0.4	-410.0	-170.0	uA	1-3
H15	-W	75	0.4	-800.0	-60.0	uA	1-3
H16	-CS	77	0.4	-800.0	-60.0	uA	1-3

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NOTES:

1. Input current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	H1-H16
Power supply current limit	0.57 A	NA	H1-H16
Forcing voltage, (Vil)	0.4 V	± 20 mV	H1-H16
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	H1-H16 H1-H16
Input current measurement	NA NA NA NA NA	± 4.4 uA ± 4.07 uA ± 4.4 uA ± 2.64 uA ± 8.14 uA	H1-H8 H9 H10 H11-H14 H15,H16

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table IX and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE X. Output voltage, high level, test.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
J1	KS0 to Q0	27	-1.0	2.4	4.5	V	1-4
J2	KS1 to Q1	28					
J3	KS2 to Q2	78					
J4	KS3 to Q3	79					
J5	KS4 to Q4	81					
J6	KS5 to Q5	83					
J7	KS6 to Q6	80					
J8	KS7 to Q7	30					
J9	KL0 to Q0	27					
J10	KL1 to Q1	28					
J11	KL2 to Q2	78					
J12	KL3 to Q3	79					
J13	KL4 to Q4	81					
J14	KL5 to Q5	83					
J15	KL6 to Q6	80					
J16	KL7 to Q7	30					
J17	KH0 to Q0	27					
J18	KH1 to Q1	28					
J19	KH2 to Q2	78					
J20	KH3 to Q3	79	-1.0	2.4	4.5	V	1-4

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TABLE X. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
J21	KH4 to Q4	81	-1.0	2.4	4.5	V	1-4
J22	KH5 to Q5	83					
J23	KH6 to Q6	80					
J24	KH7 to Q7	30					
J25	KD0 to Q0	27					
J26	KD1 to Q1	28					
J27	KD2 to Q2	78					
J28	KD3 to Q3	79					
J29	KD4 to Q4	81					
J30	KD5 to Q5	83					
J31	KD6 to Q6	80					
J32	KD7 to Q7	30	-1.0	2.4	4.5	V	1-4
J33	-RD	26	-0.38	2.5	4.5	V	1-4
J34	-SSR 0-1 OUT	6	-0.4	2.5	4.5	V	1-4

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NOTES:

1. Output voltage, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	J1-J34
Power supply current limit	0.57 A	NA	J1-J34
Forcing current, (Ioh)	- 1.0 mA - 0.38 mA - 0.4 mA	± 10.0 uA ± 3.8 uA ± 4.0 uA	J1-J32 J33 J34
Pattern voltage high, (IH) high, (VH1) low, (VL) high, (VH2) low, (VL) high, (VH)	 3.0 mA 2.4 V 0.7 V 12.0 V 0.7 V 2.0 V	 ± 100 uA ± 24 mV ± 7 mV ± 120 mV ± 7 mV ± 20 mV	 J1-J8 J9-J16 J9-J16 J17-J32 J33 J33,J34
Output voltage measurement	NA NA	± 22 mV ± 27.5 mV	J1-J32 J33,J34

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

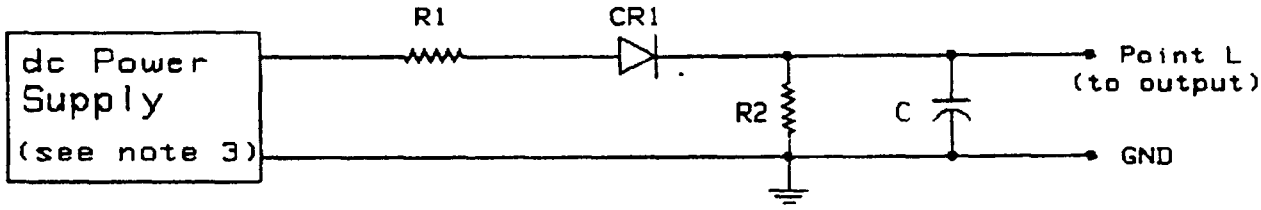
(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table X and the voltage then measured.

(e) During this test, the high and low level input threshold of every input must be verified. This may require redundant output voltage tests on some pins.

3. The load shall be as indicated on figure 6.

4. Abbreviations used are defined in 6.1.

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NOTES:

1. CR1 = 1N4148 or equivalent.

2. The load capacitance is defined as follows:

(a) C = 150 ± 15 pF (including probe and parasitic capacitance) for sample dynamic and 100 percent dynamic tests.

(b) C = 50 ± 5 pF (including probe and parasitic capacitance) for disable time test.

(c) The functional test shall be performed with only a 150 ± 15 pF load (including probe and parasitic capacitance) attached to each output.

(d) C not required for any other tests.

3. The load power supply voltage shall be 2.1 V ± 1 percent for the output disable time test and 2.5 V ± 1 percent for all other tests.

FIGURE 6. Logic load circuit.

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NOTES: - Continued.

4. Select R1 for a current flow of Iol when point L is held at Vol with the load power supply set at 2.5 V ± 1 percent. The values of Iol and Vol for each output are listed below.

5. The value of R2 for each output load shall be as specified below.

Pin number	R2 1/ (ohms)	Iol (mA)	Vol (volts)
27,28,30,78,79, 80,81,83	2400	12.0	0.4
26	6580	3.6	0.4
6	6250	4.0	0.4

1/ All resistors are metal film, ± 1 percent, 1/4 W.

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TABLE XI. Output voltage, low level, test.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
K1	KS0 to Q0	27	12.0	0.0	0.4	V	1-5
K2	KS1 to Q1	28					
K3	KS2 to Q2	78					
K4	KS3 to Q3	79					
K5	KS4 to Q4	81					
K6	KS5 to Q5	83					
K7	KS6 to Q6	80					
K8	KS7 to Q7	30					
K9	KL0 to Q0	27					
K10	KL1 to Q1	28					
K11	KL2 to Q2	78					
K12	KL3 to Q3	79					
K13	KL4 to Q4	81					
K14	KL5 to Q5	83					
K15	KL6 to Q6	80					
K16	KL7 to Q7	30					
K17	KH0 to Q0	27					
K18	KH1 to Q1	28					
K19	KH2 to Q2	78					
K20	KH3 to Q3	79	12.0	0.0	0.4	V	1-5

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TABLE XI. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
K21	KH4 to Q4	81	12.0	0.0	0.4	V	1-5
K22	KH5 to Q5	83					
K23	KH6 to Q6	80					
K24	KH7 to Q7	30					
K25	KD0 to Q0	27					
K26	KD1 to Q1	28					
K27	KD2 to Q2	78					
K28	KD3 to Q3	79					
K29	KD4 to Q4	81					
K30	KD5 to Q5	83					
K31	KD6 to Q6	80					
K32	KD7 to Q7	30	12.0	0.0	0.4	V	1-5
K33	-RD	26	3.6	0.0	0.4	V	1-5
K34	-SSR 0-1 OUT	6	4.0	0.0	0.4	V	1-5
K35	-SSR 0-1 OUT	6	4.0	0.0	0.4	V	1-5

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NOTES:

1. Output voltage, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	K1-K35
Power supply current limit	0.57 A	NA	K1-K35
Forcing current, (Iol)	12.0 mA 3.6 mA 4.0 mA	± 120.0 uA ± 36.0 uA ± 40.0 uA	K1-K32 K33 K34,K35
Pattern voltage low, (IL) low, (VL) low, (VL2) low, (VL) high, (VH) low, (VL1) high, (VH) low, (VL)	 0.9 mA 0.7 V 4.0 V 0.7 V 2.0 V 0.5 V 2.0 V 0.7 V	 ± 10 uA ± 7 mV ± 40 mV ± 7 mV ± 20 mV ± 5 mV ± 20 mV ± 7 mV	 K1-K8 K9-K16 K17-K32 K33 K33,K34 K34 K35 K35
Output voltage measurement	NA NA	± 22 mV ± 27.5 mV	K1-K32 K33-K35

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table XI and the voltage then measured.

(e) During this test, the high and low level input threshold voltage of every input must be verified. This may require redundant output voltage tests on some pins.

3. Test condition K35 will be run after K34 with the low pattern voltage (VL) rising from 0.5 V directly to 0.7 V.

4. The load shall be as indicated on figure 6.

5. Abbreviations used are defined in 6.1.

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TABLE XII. Output leakage current, high level, test.

Test cond	Signal name	Pin no.	Forcing voltage (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
L1	Q0	27	2.7	-20	20	uA	1-3
L2	Q1	28					
L3	Q2	78					
L4	Q3	79					
L5	Q4	81					
L6	Q5	83					
L7	Q6	80					
L8	Q7	30	2.7	-20	20	uA	1-3

NOTES:

1. Output leakage current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	L1-L8
Power supply current limit	0.57 A	NA	L1-L8
Forcing voltage, (Voh)	2.7 V	± 27 mV	L1-L8
Pattern voltage high, (VH)	3.0 V	± 100 mV	L1-L8
low, (VL)	0.0 V	± 100 mV	L1-L8

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Output current measurement	NA	± 0.44 uA	L1-L8

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table XII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE XIII. Output leakage current, low level, test.

Test cond	Signal name	Pin no.	Vol (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
M1	Q0	27	0.4	-20	20	uA	1-3
M2	Q1	28					
M3	Q2	78					
M4	Q3	79					
M5	Q4	81					
M6	Q5	83					
M7	Q6	80					
M8	Q7	30	0.4	-20	20	uA	1-3

NOTES:

1. Output leakage current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	M1-M8
Power supply current limit	0.57 A	NA	M1-M8
Forcing voltage, (Vol)	0.4 V	± 4 mV	M1-M8
Pattern voltage high, (VH)	3.0 V	± 100 mV	M1-M8
Pattern voltage low, (VL)	0.0 V	± 100 mV	M1-M8

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Output current measurement	NA	± 0.44 uA	M1-M8

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table XIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE XIV. Power supply current test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
N1	Iccl	1,31	200	380	mA	1-4
N2	Icch	1,31	200	380	mA	1-4

NOTES:

1. Power supply current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	N1,N2
Power supply current limit	0.57 A	NA	N1,N2
Pattern voltage high, (VH)	3.0 V	± 100 mV	N1,N2
low, (VL)	0.0 V	± 100 mV	N1,N2
Power supply current measurement	NA	± 9 mA	N1,N2

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) Pattern voltage applies to the D0-D7 inputs.

3. Test measurement to be made at t=125 mS (minimum) after the applied pattern.

4. Abbreviations used are defined in 6.1.

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TABLE XV. Functional test.

Test cond	Test description	Notes
P1	Functional test minimum power supply voltage maximum pattern rate	1-3
P2	Functional test maximum power supply voltage maximum pattern rate	1-3

NOTES:

1. Functional test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	P1 P2
Power supply current limit	0.57 A	NA	P1,P2
Pattern voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 120 mV ± 20 mV	 P1,P2 P1,P2
Output voltage verification high, (Voh) high, (Voh) low, (Vol)	 NA NA NA NA	 ± 24 mV ± 35 mV ± 5 mV ± 17 mV	 P1 P2 P1 P2

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

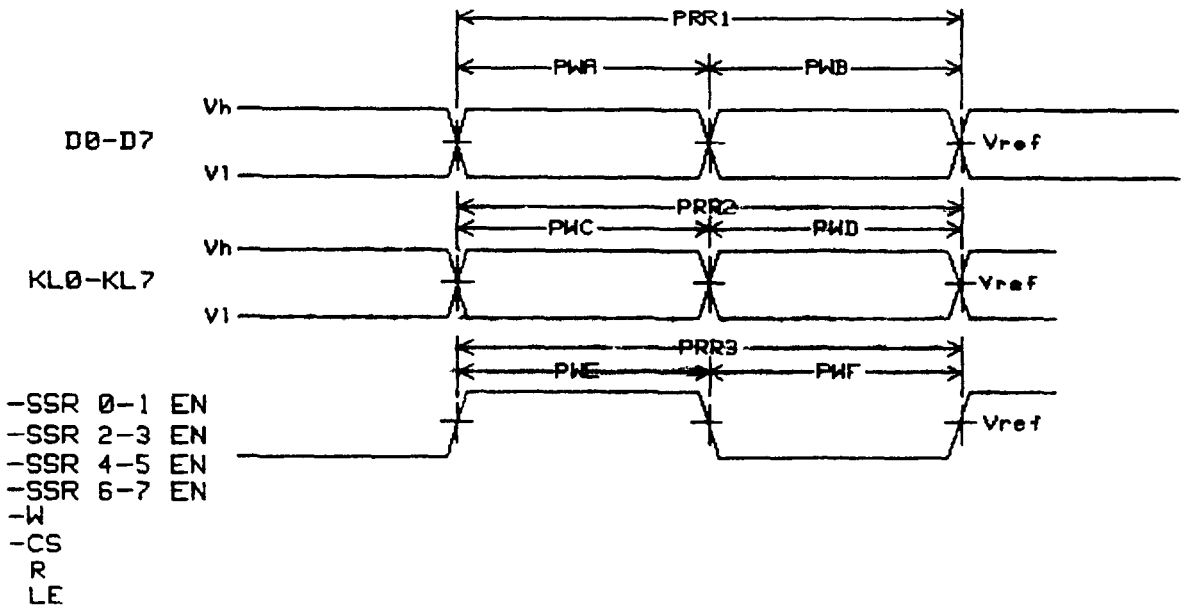
(b) Use the appropriate pattern to verify the functional performance of the module as indicated in table XV using only a capacitive load as specified on figure 6. Verify the output patterns by monitoring the outputs and comparing the results with the guaranteed output low and high levels specified in tables X and XI. The high level output (Vohi) for the RAx and RBx outputs shall be greater than or equal to 4 V, where x = 0 to 7. Wait 60 mS (minimum) before making an output measurement on Q0-Q7. Wait 20 mS (minimum) before making an output measurement on RA0, through RA7 and RB7.

(c) The module inputs shall not violate the minimum input timing parameters referenced on figure 7.

(d) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

3. Abbreviations used are defined in 6.1.

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NOTES:

1. Input parameters: PRR1 = 100 mS minimum, PRR2 = 120 mS minimum, PRR3 = 30 mS minimum, PWA = 50 mS minimum, PWB = 50 mS minimum, PWC = 60 mS minimum, PWD = 60 mS minimum, PWE = 15 mS minimum, PWF = 15 mS minimum.

2. Vref = 1.3 $\pm$ 0.1 V.

3. Pattern voltages shall be as specified in the test tolerance table.

4. Rise times (as measured between the 10 percent and 90 percent points) shall be 15 nS maximum.

5. Fall times (as measured between the 90 percent and 10 percent points) shall be 6 nS maximum.

FIGURE 7. Minimum input timing requirements for table XV.

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TABLE XVI. Dynamic test.

Test cond	Test description	Notes
Q1	Dynamic test, sample maximum pattern rate minimum power supply	1-7
Q2	Dynamic test, sample maximum pattern rate maximum power supply	1-7

NOTES:

1. Dynamic test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	Q1 Q2
Power supply current limit	0.57 A	NA	Q1,Q2
Input voltage/current			
high (VH)	2.4 V	± 24 mV	Q1,Q2
high (VH2)	12.0 V	± 120 mV	Q1,Q2 (35-42)
high (IH)	3.0 mA	± 100 uA	Q1,Q2 (43-50)
high (IH1)	6.0 mA	± 100 uA	Q1,Q2 (43-50)
low (VL)	0.4 V	± 7 mV	Q1,Q2
low (VL2)	4.0 V	± 40 mV	Q1,Q2 (35-42)
low (IL)	0.9 mA	± 10 uA	Q1,Q2 (43-50)

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Output voltage verification			
high, (Voh)	NA	± 27.5 mV	Q1
high, (Voh)	NA	± 45 mV	Q2
low, (Vol)	NA	± 5.5 mV	Q1
	NA	± 5.5 mV	Q2

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) The output compare levels shall be as follows:

High level: $Vref \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq Vref$

(c) The dynamic parameters are found in table XVII.

(d) Use the appropriate pattern to precondition the module under test and verify the output patterns by monitoring the outputs and comparing the results with the guaranteed output load and high levels specified in tables X and XI.

3. The load circuit for this test is specified on figure 6.

4. All of the input waveforms shall meet the requirements shown on figure 8.

5. This test shall be performed as indicated in table XVI.

6. Parameter numbers 1 through 16 and 35 through 50 in table XVII are 100 percent dynamic tests. Parameter numbers 17 through 34 and 51 through 72 in table XVII are sample dynamic tests.

7. Abbreviations used are defined in 6.1.

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TABLE XVII. Dynamic parameters.

Parameter number	Parameter	0 to 60°C		25 ± 5°C Initial limits		Units	Fig
		Min	Max	Min	Max		
1-8	tPLH D0-D7 inputs to (RA0,RC0)-(RA7,RC7) outputs	0.0	10.0	0.0	10.0	mS	8a
9-16	tPHL D0-D7 inputs to (RA0,RC0)-(RA7,RC7) outputs	0.0	10.0	0.0	10.0	mS	8a
17,18	tPLH -SSR 0-1 EN input to (RA0,RC0) and (RA1, RC1) outputs	0.0	10.0	0.0	10.0	mS	8b
19,20	tPHL -SSR 0-1 EN input to (RA0,RC0) and (RA1, RC1) outputs						
21,22	tPLH -SSR 2-3 EN input to (RA2,RC2) and (RA3, RC3) outputs						
23,24	tPHL -SSR 2-3 EN input to (RA2,RC2) and (RA3, RC3) outputs						
25,26	tPLH -SSR 4-5 EN input to (RA4,RC4) and (RA5, RC5) outputs	0.0	10.0	0.0	10.0	mS	8b

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TABLE XVII. Dynamic parameters - Continued.

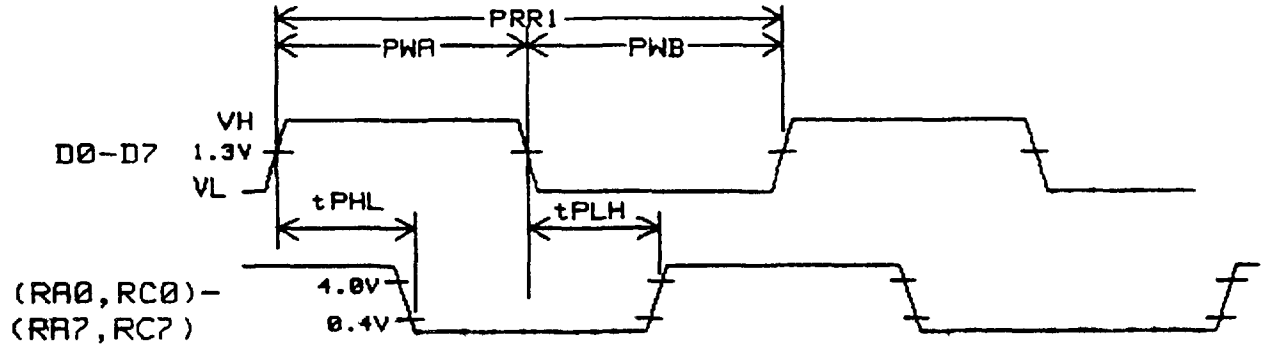
Parameter number	Parameter	0 t 60°C		25 ± 5°C Initial limits		Units	Fig
		Min	Max	Min	Max		
27,28	tPHL -SSR 4-5 EN input to (RA4,RC4) and (RA5, RC5) outputs	0.0	10.0	0.0	10.0	mS	8b
29,30	tPLH -SSR 6-7 EN input to (RA6,RC6) and (RA7, RC7) outputs						
31,32	tPHL -SSR 6-7 EN input to (RA6,RC6) and (RA7, RC7) outputs	0.0	10.0	0.0	10.0	mS	8b
33	tPLH -SSR 0-1 EN input to -SSR 0-1 OUT output	10.0	104.0	10.0	104.0	nS	8c
34	tPHL -SSR 0-1 EN input to -SSR 0-1 OUT output	10.0	104.0	10.0	104.0	nS	8c
35-42	tPLH KH0-KH7 inputs to Q0-Q7 outputs	0.0	55.0	0.0	55.0	mS	8d,8e 8h,8i
43-50	tPHL KH0-KH7 inputs to Q0-Q7 outputs	5.0	55.0	5.0	55.0	mS	8f-8g 8j,8k
51	tPLH -W input to -RD output	6.0	90.0	6.0	90.0	nS	8l

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TABLE XVII. Dynamic parameters - Continued.

Parameter number	Parameter	0 to 60°C		25 ± 5°C Initial limits		Units	Fig
		Min	Max	Min	Max		
52	tPHL -W input to -RD output	6.0	90.0	6.0	90.0	nS	8l
53	tPLH -CS input to -RD output	4.0	60.0	4.0	60.0	nS	8l
54	tPHL -CS input to -RD output	4.0	60.0	4.0	60.0	nS	8l
55	tPLH R input to -RD output	2.0	30.0	2.0	30.0	nS	8l
56	tPHL R input to -RD output	2.0	30.0	2.0	30.0	nS	8l
57-64	tPZH tri-state to high level outputs on Q0-Q7	5.0	67.0	5.0	67.0	nS	8m
65-72	tPZL tri-state to low level outputs on Q0-Q7	5.0	77.0	5.0	77.0	nS	8m

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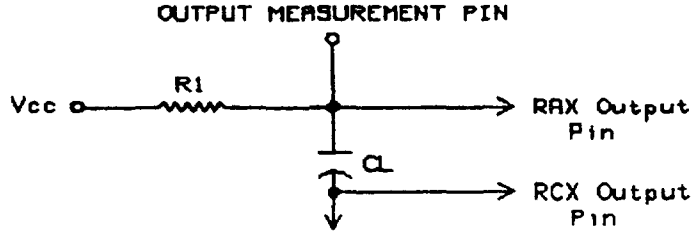


NOTES:

1. Input parameters: PRR1 = 1 S (minimum), PWA = 500 mS (minimum), PWB = 500 mS (minimum), tR = tF $\leq$ 20 nS (10 percent to 90 percent).

2. -SSR 0-1 EN, -SSR 2-3 EN, -SSR 4-5 EN, and -SSR 6-7 EN should be connected to ground.

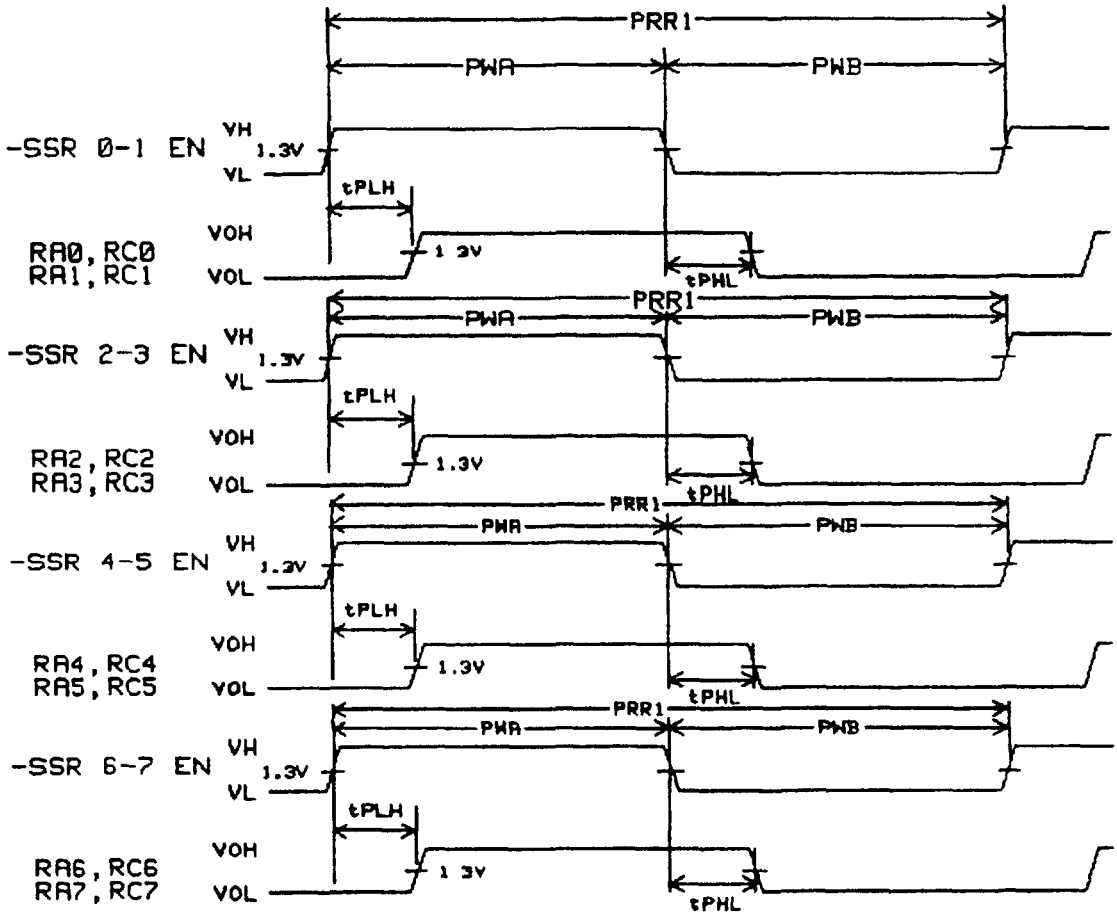
3. Use the following output load circuit for RAX and RCX outputs (where X = 0 to 7).



CL = 150 pF (including jig and probe capacitance)
R1 = 50 Ohm $\pm$ 5 percent.

FIGURE 8a. Dynamic test waveforms for tables XVI and XVII.

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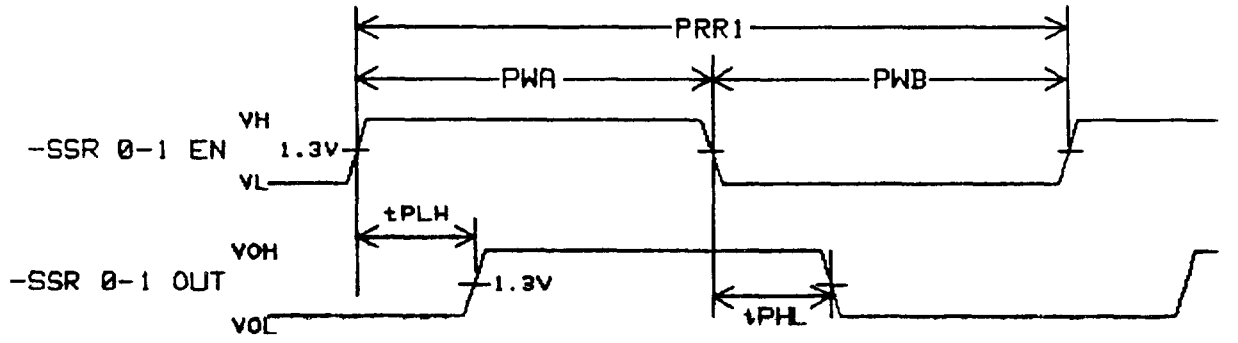
NOTE:

1. Input parameters:

   PRR1 = 1 S minimum.
   PWA = PWB = 0.5 S minimum.
   tR = tF ≤ 20 nS (10 percent to 90 percent).

FIGURE 8b. Dynamic test waveforms for tables XVI and XVII.

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NOTE:

1. Input parameters:

   PRR1 = 1 uS minimum.
   PWA = PWB = 0.5 uS minimum.
   tR ≤ 15 nS, tF ≤ 6 nS (10 percent to 90 percent).

FIGURE 8c. Dynamic test waveforms for tables XVI and XVII.

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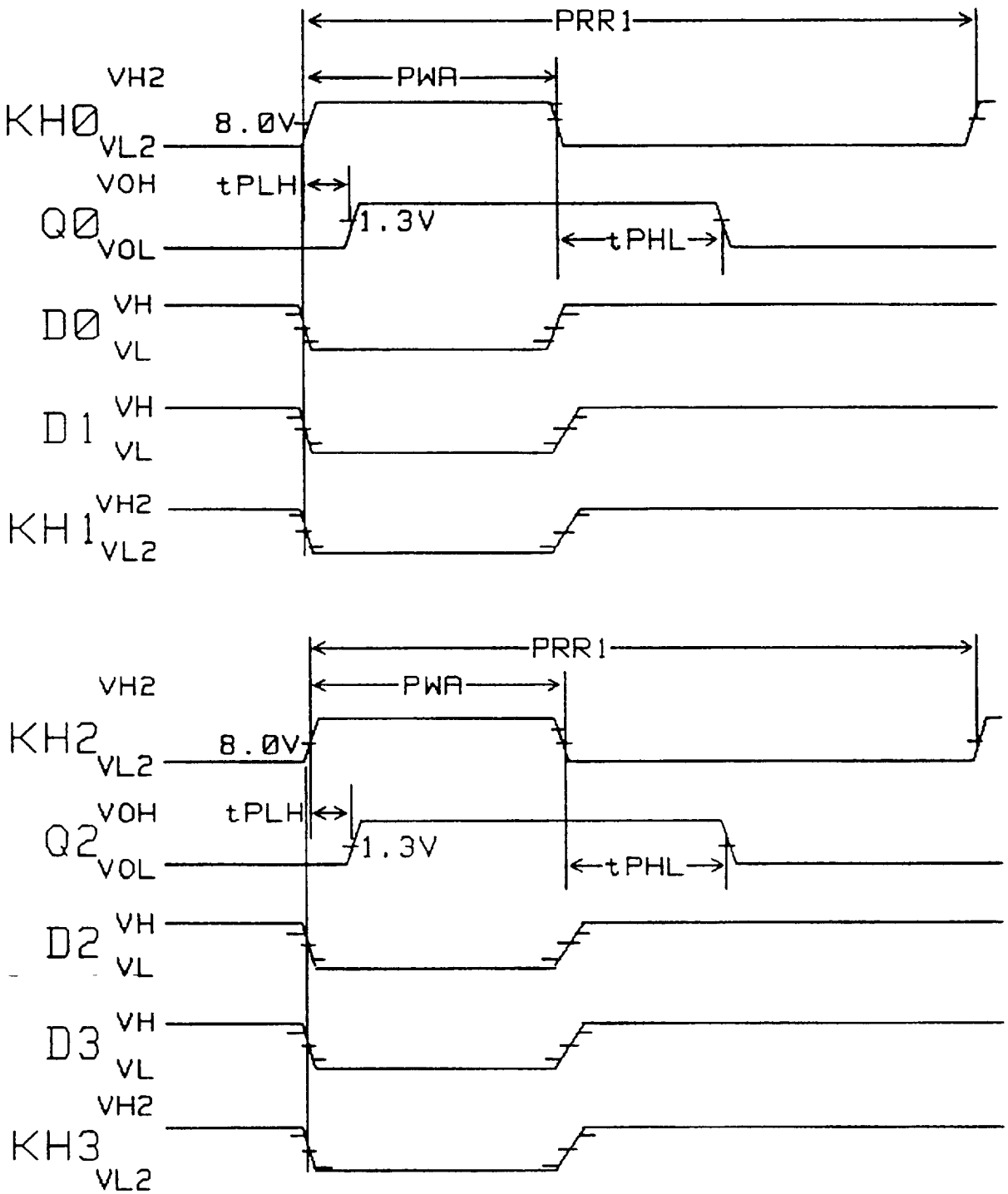


FIGURE 8d. Dynamic test waveforms for tables XVI and XVII.

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Check the source to verify that this is the current version before use.

NOTES:

1. Input parameters: PRR1 = 1 S (minimum), PWA = 100 mS (minimum), tR = tF $\leq$ 20 ns (10 percent to 90 percent).

2. -SSR 0-1 EN, -SSR 2-3 EN, -SSR 4-5 EN and -SSR 6-7 EN should be connected to ground.

3. Test parameters 35 through 42 of the propagation delay measurements must be measured with the input vector pattern conditions as specified below:

KH (X+1)	D (X)	D (X+1)	
VH2	VH	VH	Setup test vector 1
VL2	VL	VL	Test measurement 1

(a) The setup test vectors must be performed in order to condition the module. No output measurement is required during this test.

(b) Where: X = 0 to 7.

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Check the source to verify that this is the current version before use.

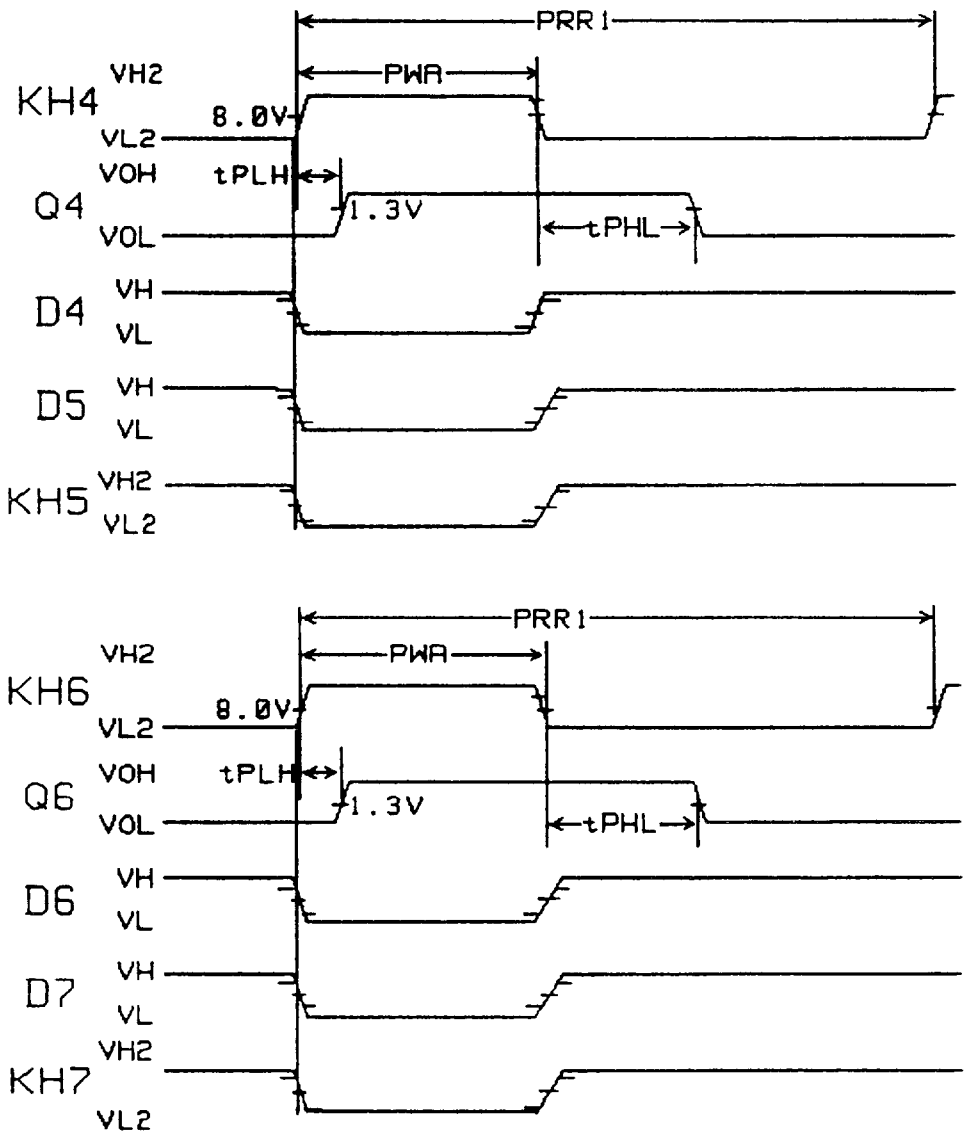


FIGURE 8e. Dynamic test waveforms for tables XVI and XVII.

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Check the source to verify that this is the current version before use.

NOTES:

1. Input parameters: PRR1 = 1 S (minimum), PWA = 100 mS (minimum), tR = tF $\leq$ 20 ns (10 percent to 90 percent).

2. -SSR 0-1 EN, -SSR 2-3 EN, -SSR 4-5 EN and -SSR 6-7 EN should be connected to ground.

3. Test parameters 35 through 42 of the propagation delay measurements must be measured with the input vector pattern conditions as specified below:

KH (X+1)	D (X)	D (X+1)	
VH2	VH	VH	Setup test vector 1
VL2	VL	VL	Test measurement 1

(a) The setup test vectors must be performed in order to condition the module. No output measurement is required during this test.

(b) Where: X = 0 to 7.

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Check the source to verify that this is the current version before use.

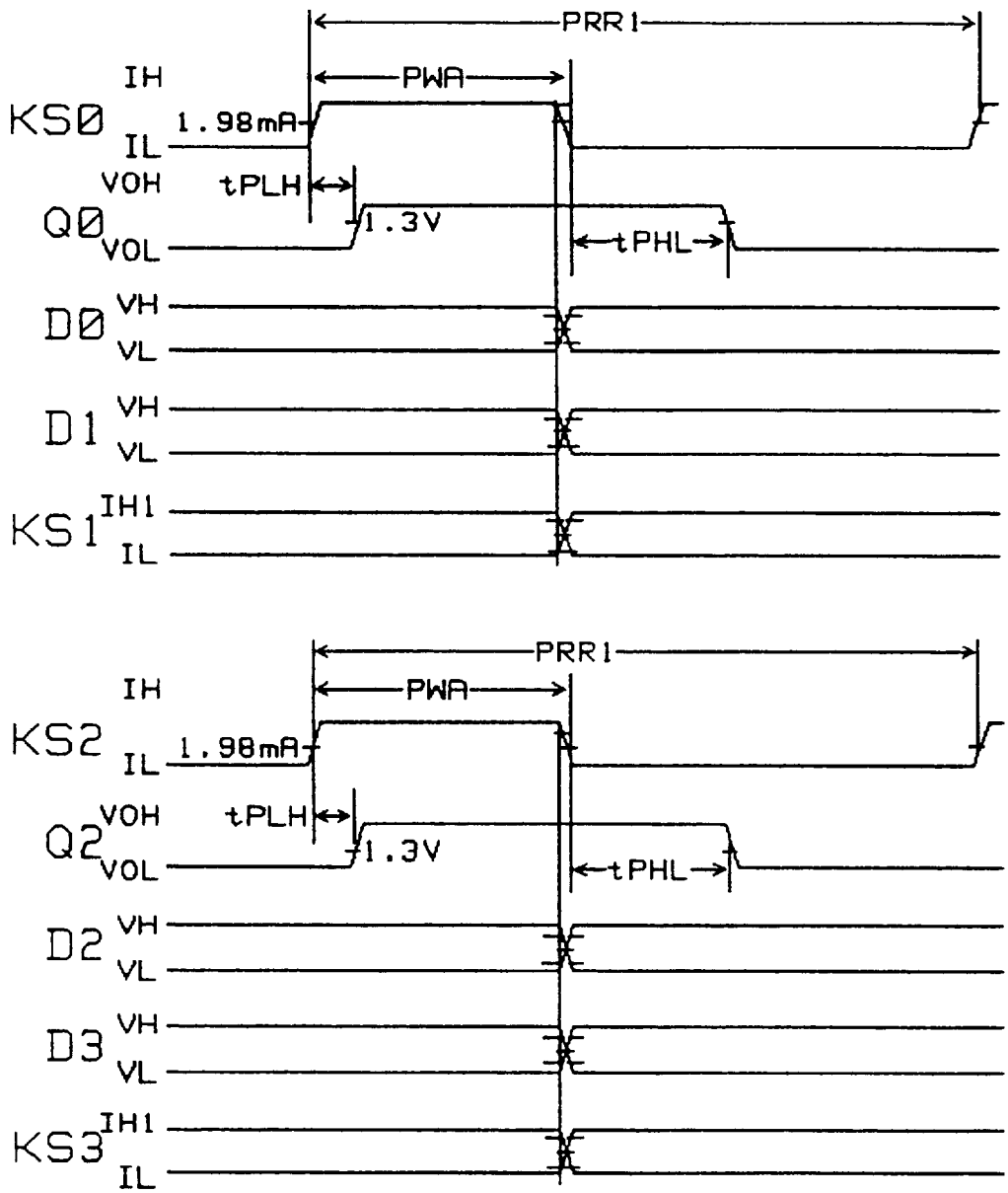


FIGURE 8f. Dynamic test waveforms for tables XVI and XVII.

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Check the source to verify that this is the current version before use.

NOTES:

1. Input parameters: PRR1 = 1 S (minimum), PWA = 100 mS (minimum), tR = tF $\leq$ 20 ns (10 percent to 90 percent).

2. -SSR 0-1 EN, -SSR 2-3 EN, -SSR 4-5 EN and -SSR 6-7 EN should be connected to ground.

3. Test parameters 43 through 50 of the propagation delay measurements must be measured with eight of the twelve input vector pattern conditions as specified below:

KS (X+1)	D (X)	D (X+1)	
IL	VL	VL	Setup test vector 1
IH	VH	VH	Test measurement 1
IL	VL	VL	Test measurement 2
IL	VH	VL	Setup test vector 2
IH	VL	VH	Test measurement 3
IL	VH	VL	Test measurement 4
IL	VL	VH	Setup test vector 3
IH	VH	VL	Test measurement 5
IL	VL	VH	Test measurement 6
IL	VH	VH	Setup test vector 4
IH	VL	VL	Test measurement 7
IL	VH	VH	Test measurement 8

(a) The setup test vectors must be performed in order to condition the module. No output measurement is required during this test.

(b) Where: X = 0 to 7.

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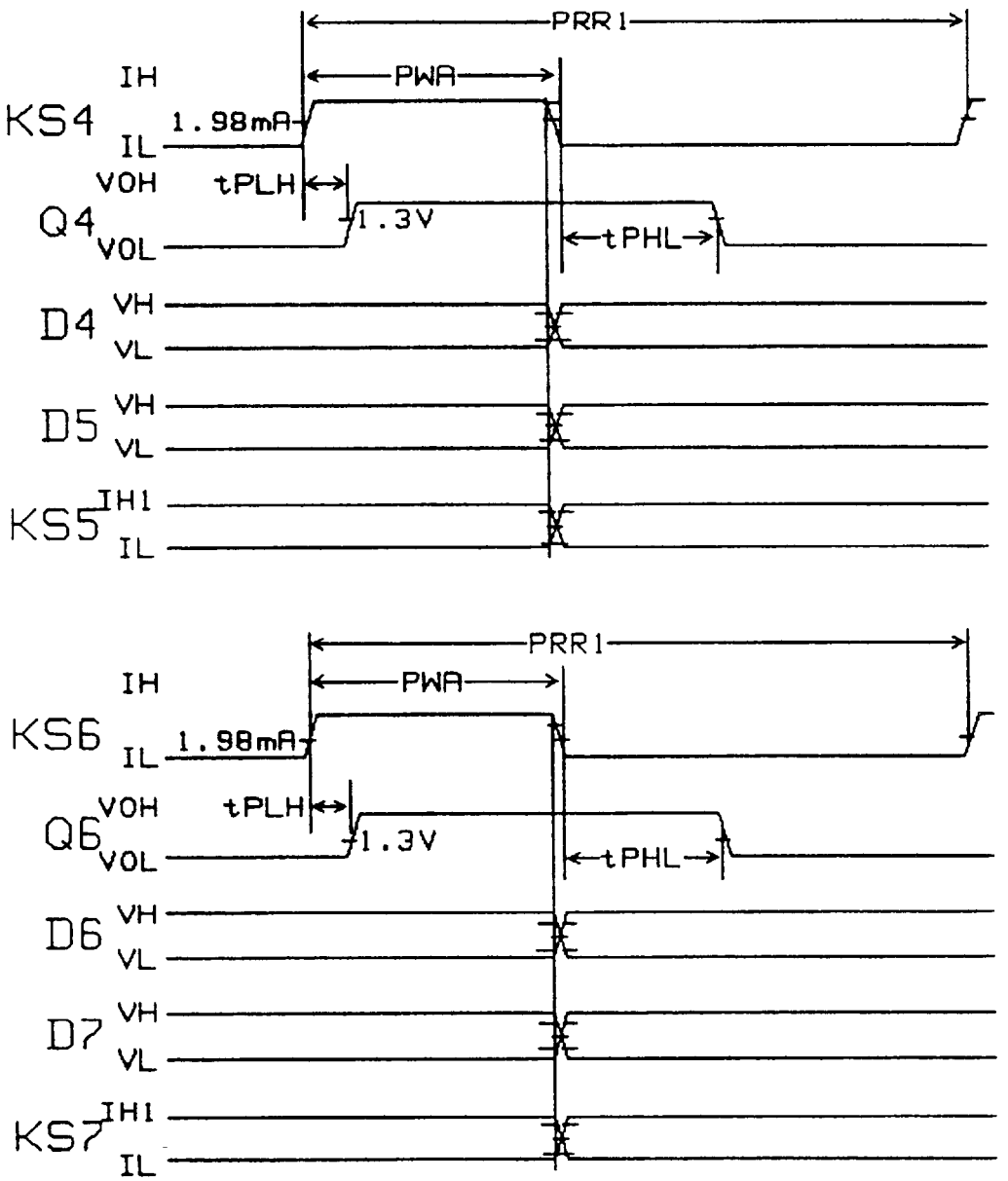


FIGURE 8g. Dynamic test waveforms for tables XVI and XVII.

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Check the source to verify that this is the current version before use.

NOTES:

1. Input parameters: PRR1 = 1 S (minimum), PWA = 100 mS (minimum), tR = tF $\leq$ 20 ns (10 percent to 90 percent).

2. -SSR 0-1 EN, -SSR 2-3 EN, -SSR 4-5 EN and -SSR 6-7 EN should be connected to ground.

3. Test parameters 35 through 42 of the propagation delay measurements must be measured with eight of the twelve input vector pattern conditions as specified below:

KS (X+1)	D (X)	D (X+1)	
IL	VL	VL	Setup test vector 1
IH	VH	VH	Test measurement 1
IL	VL	VL	Test measurement 2
IL	VH	VL	Setup test vector 2
IH	VL	VH	Test measurement 3
IL	VH	VL	Test measurement 4
IL	VL	VH	Setup test vector 3
IH	VH	VL	Test measurement 5
IL	VL	VH	Test measurement 6
IL	VH	VH	Setup test vector 4
IH	VL	VL	Test measurement 7
IL	VH	VH	Test measurement 8

(a) The setup test vectors must be performed in order to condition the module. No output measurement is required during this test.

(b) Where: X = 0 to 7.

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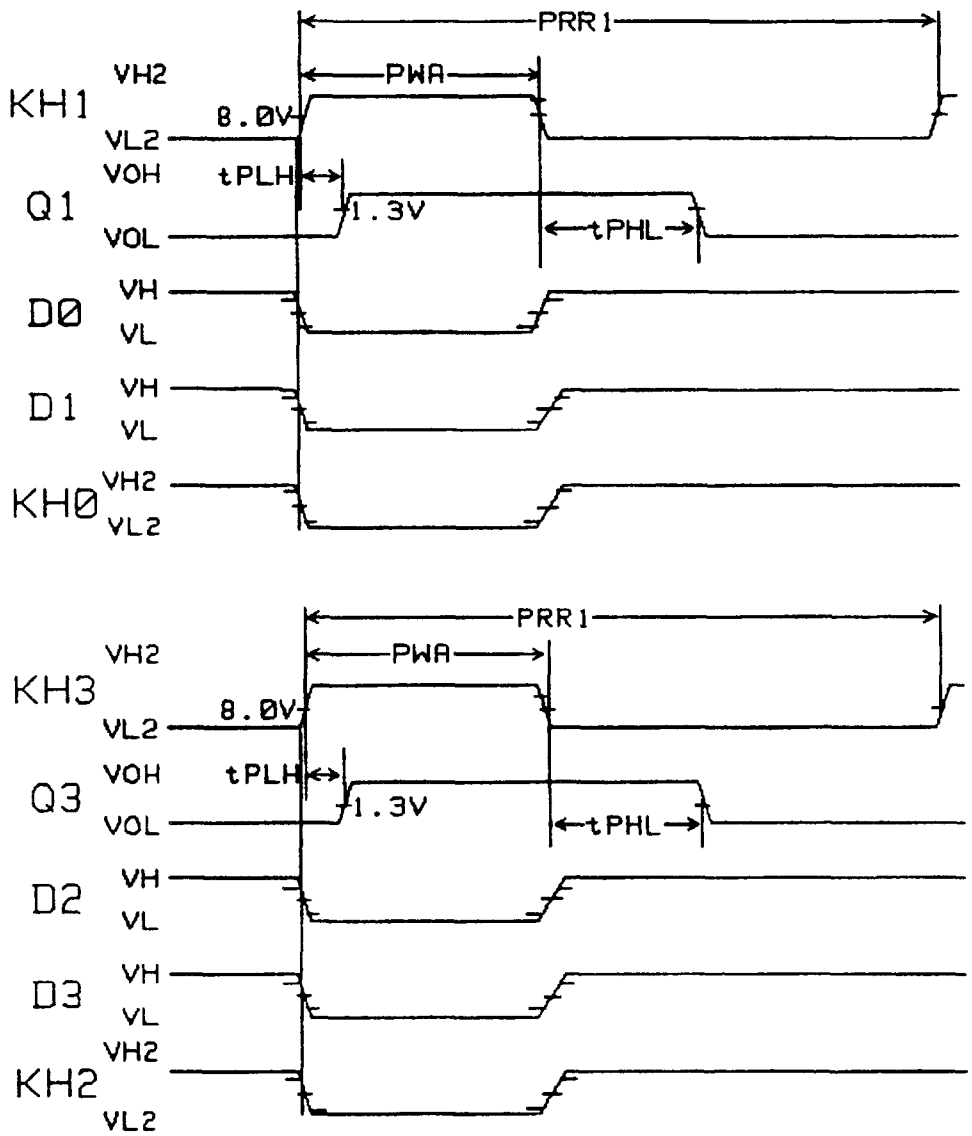


FIGURE 8h. Dynamic test waveforms for tables XVI and XVII.

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Check the source to verify that this is the current version before use.

NOTES:

1. Input parameters: PRR1 = 1 S (minimum), PWA = 100 mS (minimum), tR = tF $\leq$ 20 ns (10 percent to 90 percent).

2. -SSR 0-1 EN, -SSR 2-3 EN, -SSR 4-5 EN and -SSR 6-7 EN should be connected to ground.

3. Test parameters 35 through 42 of the propagation delay measurements must be measured with the input vector pattern conditions as specified below:

KH (X-1)	D (X)	D (X-1)	
VH2	VH	VH	Setup test vector 1
VL2	VL	VL	Test measurement 1

(a) The setup test vectors must be performed in order to condition the module. No output measurement is required during this test.

(b) Where: X = 0 to 7.

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Check the source to verify that this is the current version before use.

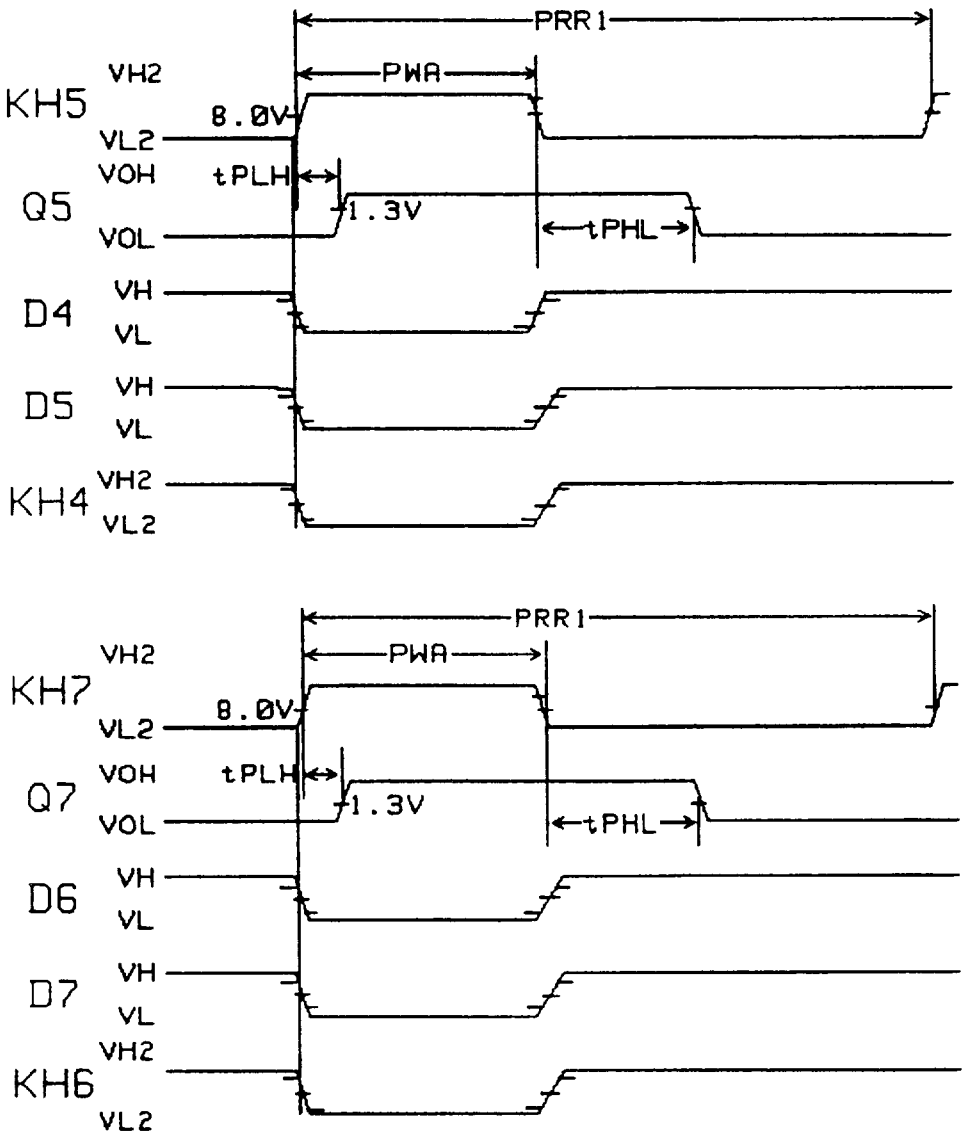


FIGURE 8i. Dynamic test waveforms for tables XVI and XVII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:10Z
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NOTES:

1. Input parameters: PRR1 = 1 S (minimum), PWA = 100 mS (minimum), tR = tF $\leq$ 20 ns (10 percent to 90 percent).

2. -SSR 0-1 EN, -SSR 2-3 EN, -SSR 4-5 EN and -SSR 6-7 EN should be connected to ground.

3. Test parameters 35 through 42 of the propagation delay measurements must be measured with the input vector pattern conditions as specified below:

KH (X-1)	D (X)	D (X-1)	
VH2	VH	VH	Setup test vector 1
VL2	VL	VL	Test measurement 1

(a) The setup test vectors must be performed in order to condition the module. No output measurement is required during this test.

(b) Where: X = 0 to 7.

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Check the source to verify that this is the current version before use.

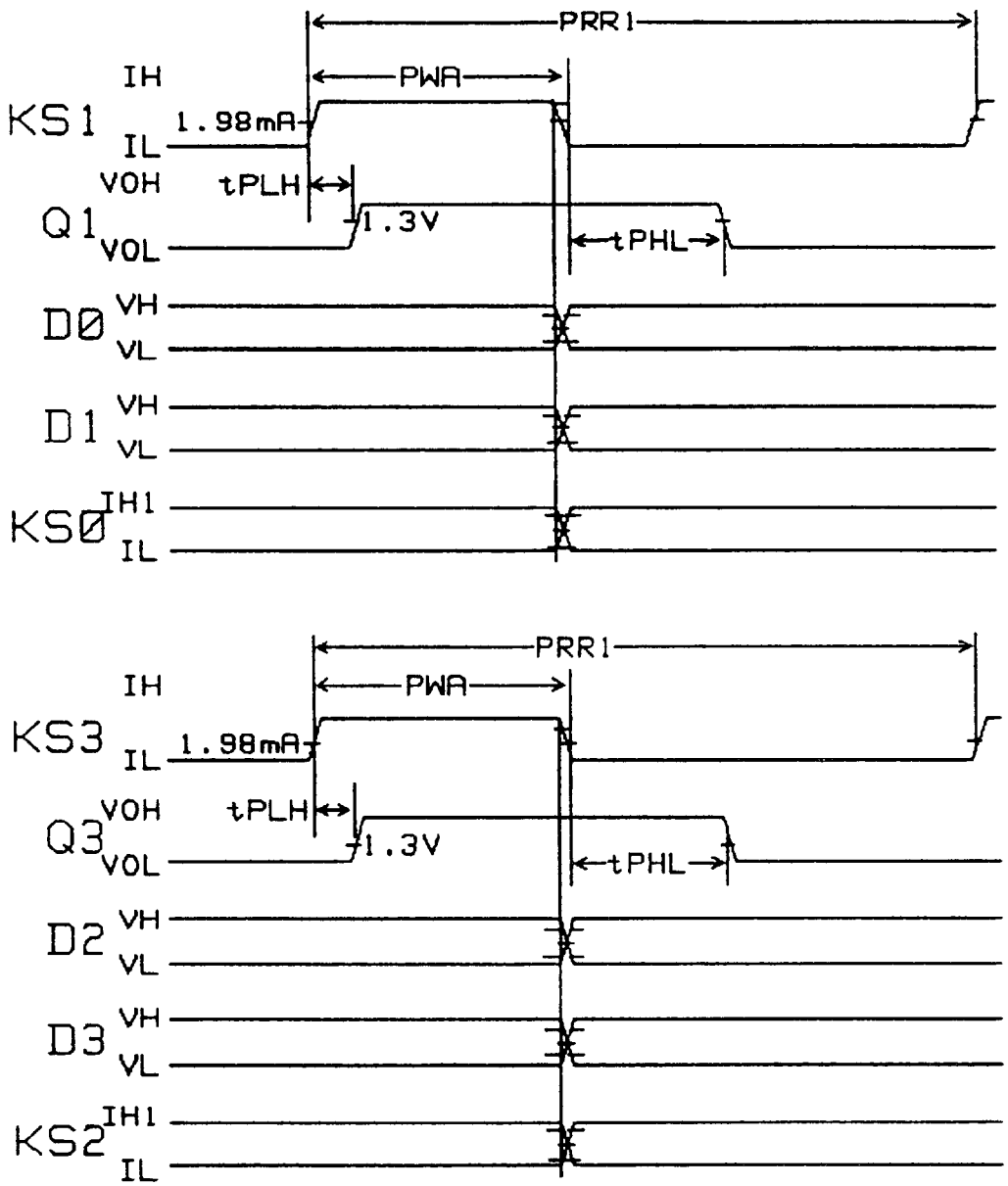


FIGURE 8j. Dynamic test waveforms for tables XVI and XVII.

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NOTES:

1. Input parameters: PRR1 = 1 S (minimum), PWA = 100 mS (minimum), tR = tF $\leq$ 20 ns (10 percent to 90 percent).

2. -SSR 0-1 EN, -SSR 2-3 EN, -SSR 4-5 EN and -SSR 6-7 EN should be connected to ground.

3. Test parameters 43 through 50 of the propagation delay measurements must be measured with eight of the twelve input vector pattern conditions as specified below:

KS (X-1)	D (X)	D (X-1)	
IL	VL	VL	Setup test vector 1
IH	VH	VH	Test measurement 1
IL	VL	VL	Test measurement 2
IL	VH	VL	Setup test vector 2
IH	VL	VH	Test measurement 3
IL	VH	VL	Test measurement 4
IL	VL	VH	Setup test vector 3
IH	VH	VL	Test measurement 5
IL	VL	VH	Test measurement 6
IL	VH	VH	Setup test vector 4
IH	VL	VL	Test measurement 7
IL	VH	VH	Test measurement 8

(a) The setup test vectors must be performed in order to condition the module. No output measurement is required during this test.

(b) Where: X = 0 to 7.

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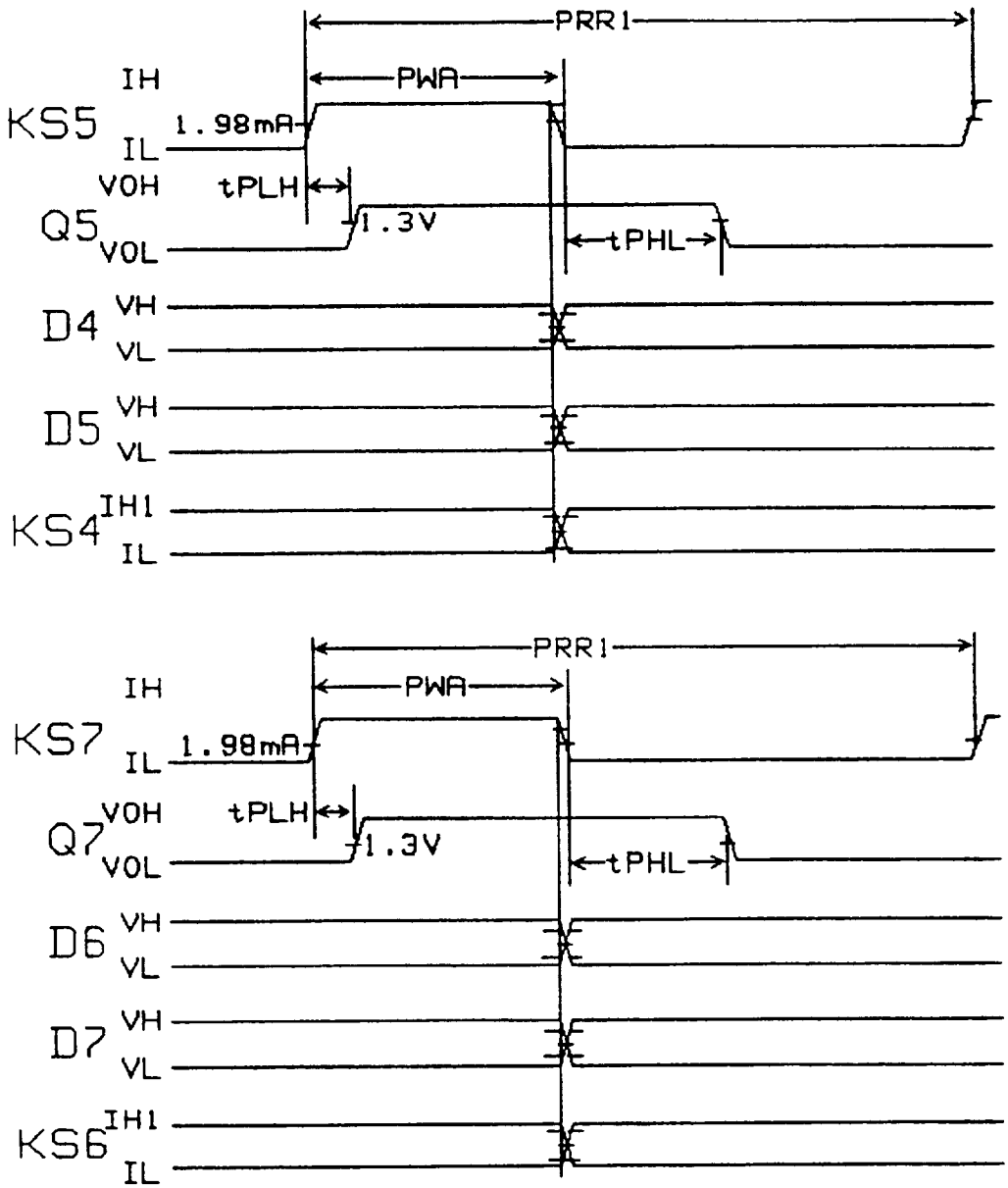


FIGURE 8k. Dynamic test waveforms for tables XVI and XVII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:10Z
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NOTES:

1. Input parameters: PRR1 = 1 S (minimum), PWA = 100 mS (minimum), tR = tF $\leq$ 20 ns (10 percent to 90 percent).

2. -SSR 0-1 EN, -SSR 2-3 EN, -SSR 4-5 EN and -SSR 6-7 EN should be connected to ground.

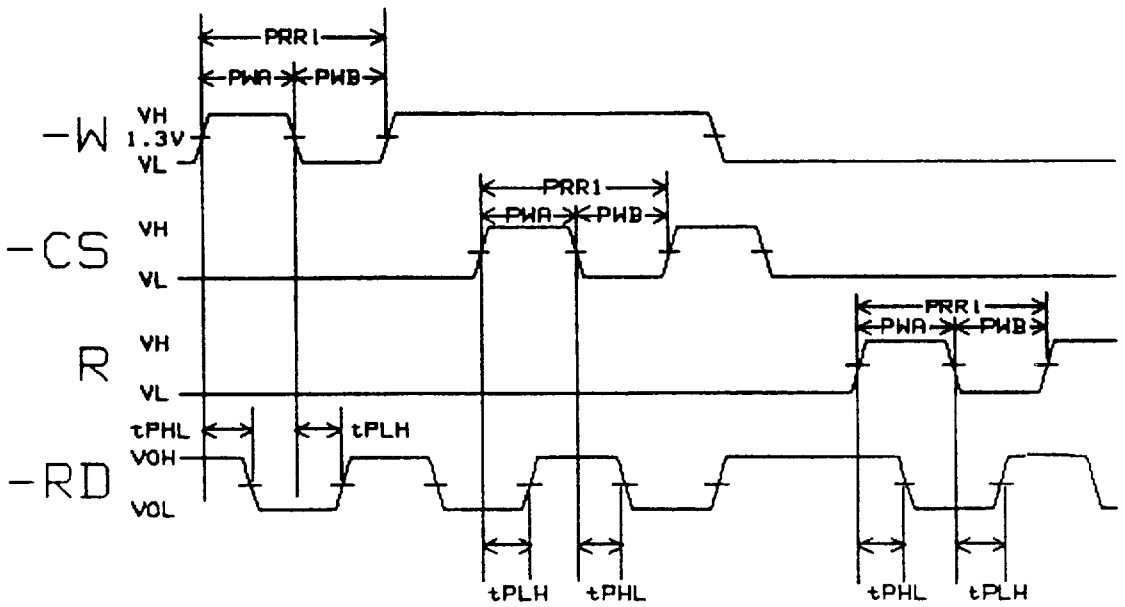
3. Test parameters 43 through 50 of the propagation delay measurements must be measured with eight of the twelve input vector pattern conditions as specified below:

KS (X-1)	D (X)	D (X-1)	
IL	VL	VL	Setup test vector 1
IH	VH	VH	Test measurement 1
IL	VL	VL	Test measurement 2
IL	VH	VL	Setup test vector 2
IH	VL	VH	Test measurement 3
IL	VH	VL	Test measurement 4
IL	VL	VH	Setup test vector 3
IH	VH	VL	Test measurement 5
IL	VL	VH	Test measurement 6
IL	VH	VH	Setup test vector 4
IH	VL	VL	Test measurement 7
IL	VH	VH	Test measurement 8

(a) The setup test vectors must be performed in order to condition the module. No output measurement is required during this test.

(b) Where: X = 0 to 7.

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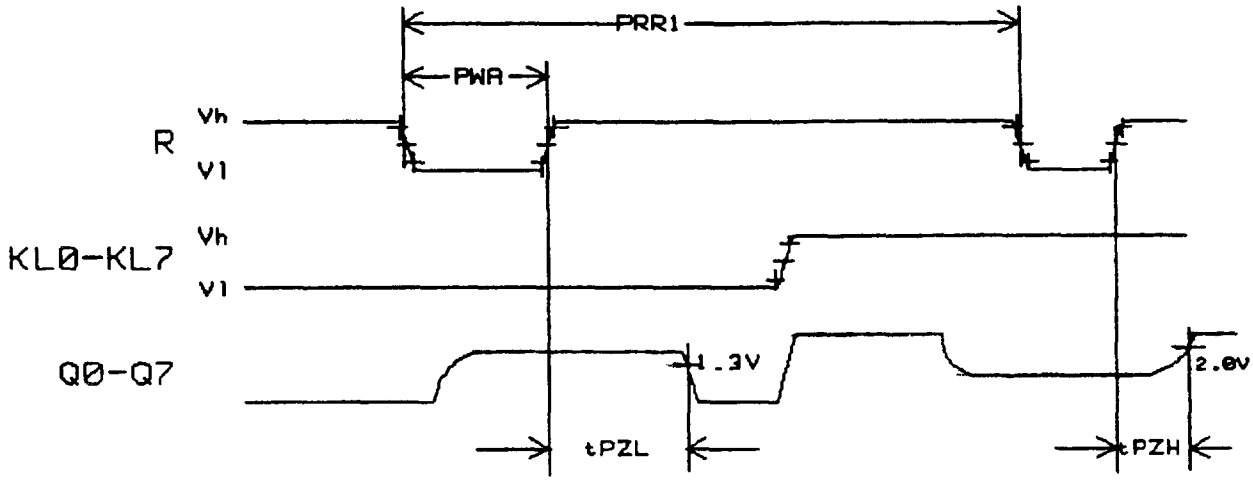
NOTE:

1. Input parameters:

   PRR1 = 2 uS minimum.
   PWA = PWB = 1 uS minimum.
   tR ≤ 15 nS, tF ≤ 6 nS (10 percent to 90 percent).

FIGURE 81. Dynamic test waveforms for tables XVI and XVII.

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NOTES:

1. Input parameters:

   PRR1 = 110 mS minimum.
   PWA $\geq$ 55 mS minimum.
   tR $\leq$ 15 nS, tF $\leq$ 6 nS (10 percent to 90 percent).

2. Condition -W low (VL) and LE high (VH).

FIGURE 8m. Dynamic test waveforms for tables XVI and XVII.

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TABLE XVIII. Power supply filter capacitance test.

Test cond	Parameter	Pin numbers	25 ± 5°C Initial limits Min	Max	Units	Notes
R	PSFC	1,31	1.59	2.69	uF	1-3

NOTES:

1. Power supply filter capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance bridge frequency	1 KHz	± 100 Hz	R
Capacitance bridge voltage	0.3 Vp	± 30 mVp	R
Capacitance measurement	NA	± 33 nF	R

2. The capacitance bridge used in this test shall be item C01. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XVIII.

3. Abbreviations used are defined in 6.1.

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TABLE XIX. Power supply transient amplitude test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
S	PSTA	1,31	0	400	mV	1-3

NOTES:

1. Power supply transient amplitude test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 250 mV	S
Power supply current limit	0.57 A	NA	S
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 S S
Power supply transient amplitude measurement	NA	± 4.4 mV	S

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table to an external source. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and make the indicated measurement for the test condition shown in table XIX.

(c) All Vcc pins shall be shorted together and the scope point, probe 'A', shall be connected to the pin indicated in table XIX.

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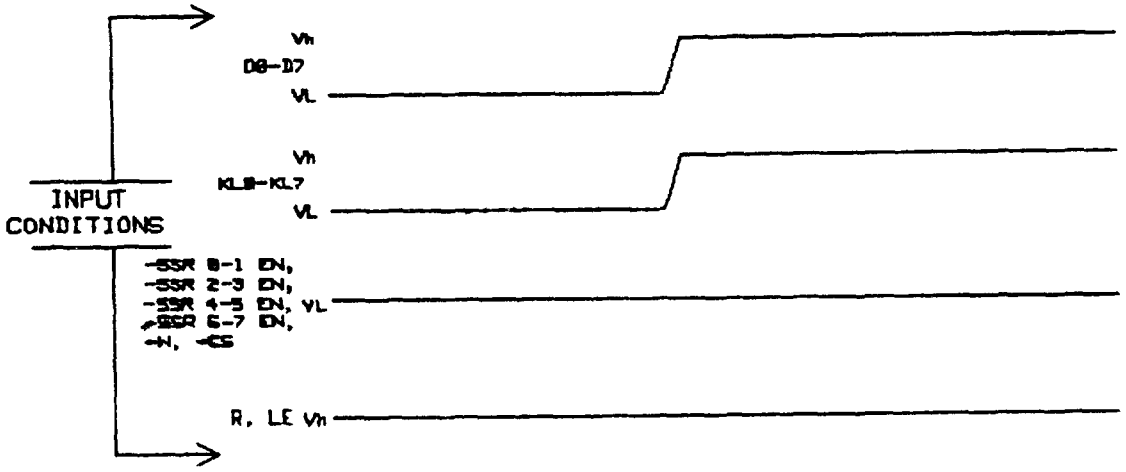
NOTES: - Continued.

(d) Using the oscilloscope, D06, inspect for the most negative spike as shown on figure 9 at the pattern line with the maximum power supply current.

(e) The maximum probe capacitance is 15 pF. There shall be no other capacitance present on Vcc to ground.

3. Abbreviations used are defined in 6.1.

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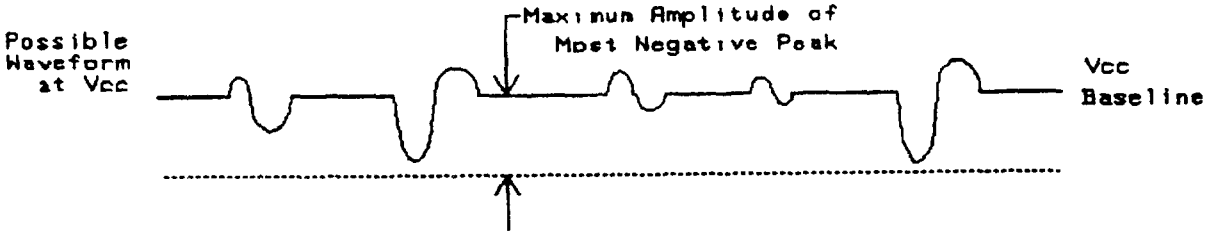


FIGURE 9. Power supply transient amplitude waveform for table XIX.

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TABLE XX. Input and output capacitance test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
T1	KS0	58	2	100	pF	1-3
T2	KS1	5				
T3	KS2	66				
T4	KS3	13				
T5	KS4	86				
T6	KS5	34				
T7	KS6	90				
T8	KS7	93				
T9	KL0	8				
T10	KL1	55				
T11	KL2	16				
T12	KL3	63				
T13	KL4	36				
T14	KL5	84				
T15	KL6	44				
T16	KL7	42				
T17	KH0	59				
T18	KH1	4				
T19	KH2	54				
T20	KH3	22	2	100	pF	1-3

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TABLE XX. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
T21	KH4	97	2	100	pF	1-3
T22	KH5	32				
T23	KH6	49				
T24	KH7	92				
T25	KD0	2				
T26	KD1	53				
T27	KD2	3				
T28	KD3	70				
T29	KD4	98				
T30	KD5	29				
T31	KD6	99				
T32	KD7	48				
T33	-SSR 0-1 EN	57				
T34	-SSR 2-3 EN	14				
T35	-SSR 4-5 EN	33				
T36	-SSR 6-7 EN	91				
T37	-W	75				
T38	R	76				
T39	-CS	77				
T40	LE	82	2	100	pF	1-3

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TABLE XX. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
T41	-SSR 0-1 OUT	6	2	100	pF	1-3
T42	-RD	26				
T43	Q0	27				
T44	Q1	28				
T45	Q2	78				
T46	Q3	79				
T47	Q4	81				
T48	Q5	83				
T49	Q6	80				
T50	Q7	30				
T51	D0	72				
T52	D1	21				
T53	D2	71				
T54	D3	20				
T55	D4	73				
T56	D5	23				
T57	D6	74				
T58	D7	24	2	100	pF	1-3

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NOTES:

1. Input and output capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance meter frequency	1 MHz	± 100 KHz	T1-T58
Capacitance meter voltage	15 mV	± 1.5 mV	T1-T58
Capacitance measurement	NA	± 5 pF	T1-T58

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XX.

3. Abbreviations used are defined in 6.1.

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TABLE XXI. Switch "ON" resistance test.

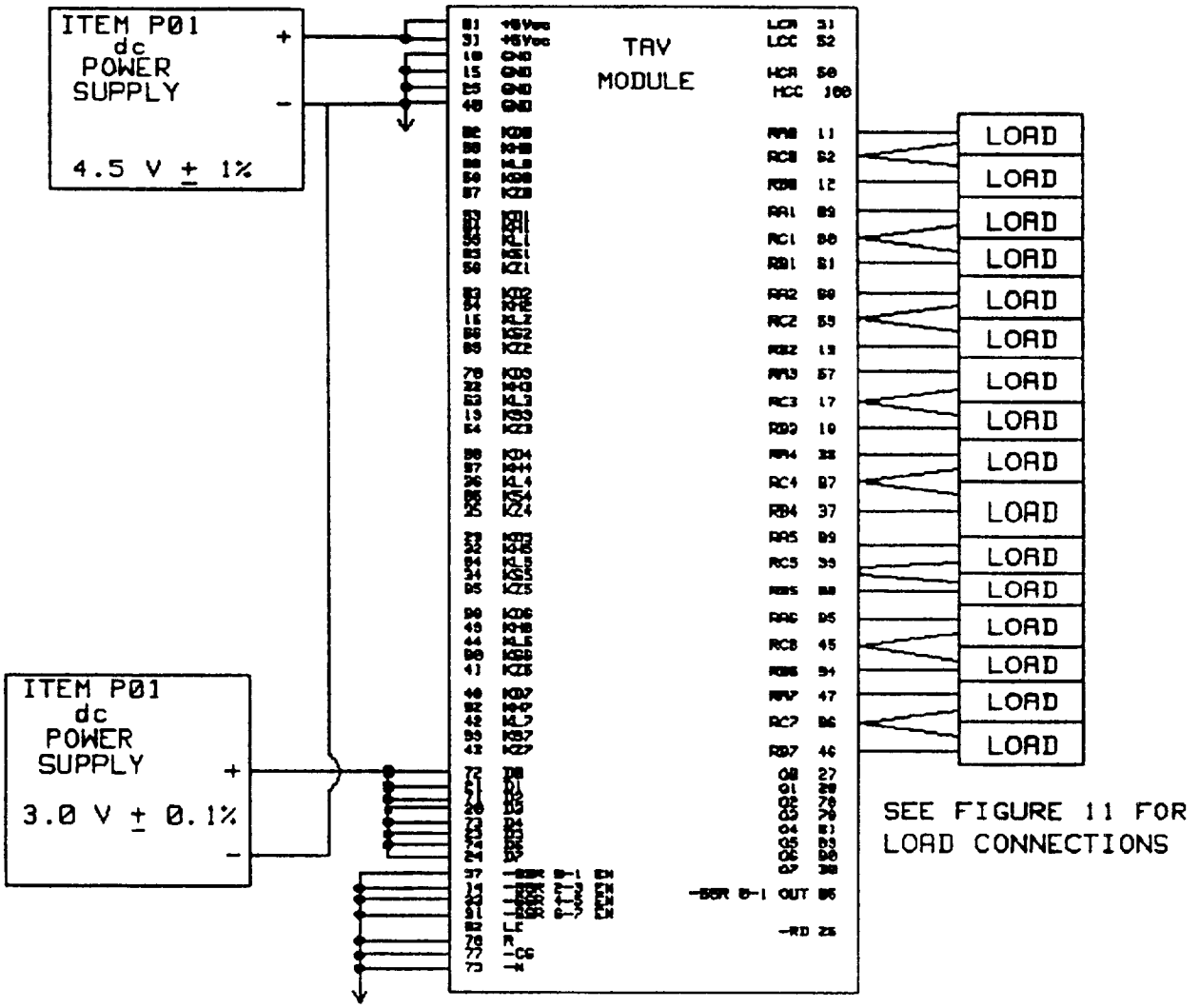
Test cond	Signal name	Pin no.	IL (A)	25 ± 5°C Initial limits Min	Max	Units	Figs
U1	RA0	11	0.25	0	145	mV	10,11
U2	RA1	9					
U3	RA2	68					
U4	RA3	67					
U5	RA4	38					
U6	RA5	89					
U7	RA6	95					
U8	RA7	47					
U9	RB0	12					
U10	RB1	61					
U11	RB2	19					
U12	RB3	18					
U13	RB4	37					
U14	RB5	88					
U15	RB6	94					
U16	RB7	46	0.25	0	145	mV	10,11
U17	RA0	11	-0.25	-145	0	mV	10,11
U18	RA1	9					
U19	RA2	68					
U20	RA3	67	-0.25	-145	0	mV	10,11

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TABLE XXI. Switch "ON" resistance test - Continued.

Test cond	Signal name	Pin no.	IL (A)	25 ± 5°C Initial limits Min	Max	Units	Figs
U21	RA4	38	-0.25	-145	0	mV	10,11
U22	RA5	89					
U23	RA6	95					
U24	RA7	47					
U25	RB0	12					
U26	RB1	61					
U27	RB2	19					
U28	RB3	18					
U29	RB4	37					
U30	RB5	88					
U31	RB6	94					
U32	RB7	46	-0.25	-145	0	mV	10,11

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NOTES:

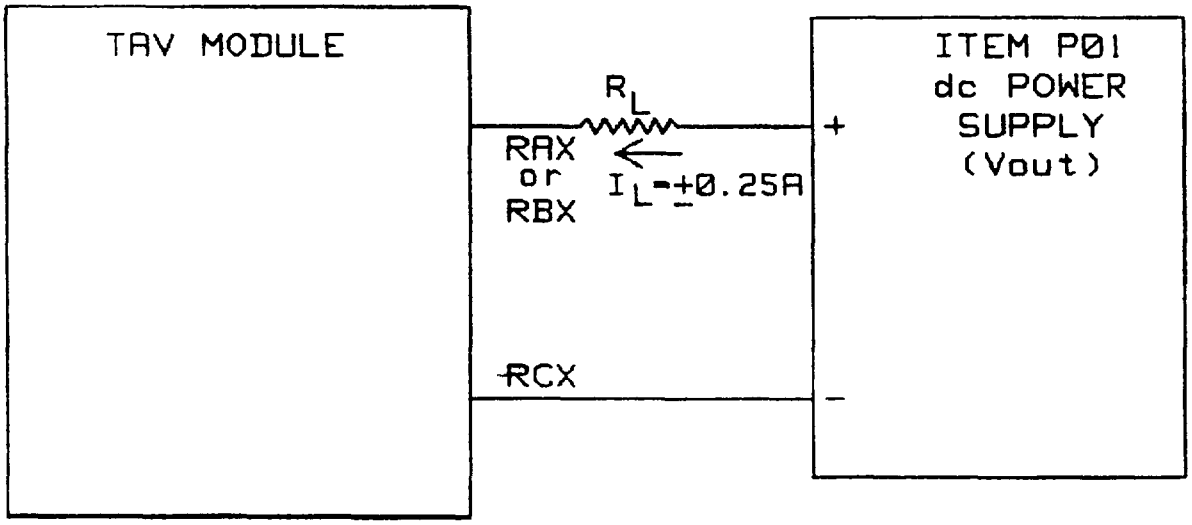
1. Apply the load to each output, one at a time.

2. Measure the voltage, Vout, across the output terminals RAX, RCX, and RBX, RCX; where X = 0 to 7.

3. Perform the calculation Ron = $\frac{\text{Vout}}{\text{IL}}$.

FIGURE 10. Switch "ON" resistance test circuit for table XXI.

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WHERE (X = 0 to 7)

SELECT R_L AS FOLLOWS:

$$R_L = \frac{Vout}{0.25A}$$

RESISTOR VALUE BASED ON THE OUTPUT VOLTAGE OF THE POWER SUPPLY

FIGURE 11. Solid state relay load circuit for table XXI.

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TABLE XXII. FET switch leakage current test.

Test cond	Signal name	A to pin	B to pin	Forcing voltage (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Fig
V1	RA0	11	62	70	0	25	uA	12
V2	RA1	9	60					
V3	RA2	68	69					
V4	RA3	67	17					
V5	RA4	38	87					
V6	RA5	89	39					
V7	RA6	95	45					
V8	RA7	47	96					
V9	RB0	12	62					
V10	RB1	61	60					
V11	RB2	19	69					
V12	RB3	18	17					
V13	RB4	37	87					
V14	RB5	88	39					
V15	RB6	94	45					
V16	RB7	46	96	70	0	25	uA	12

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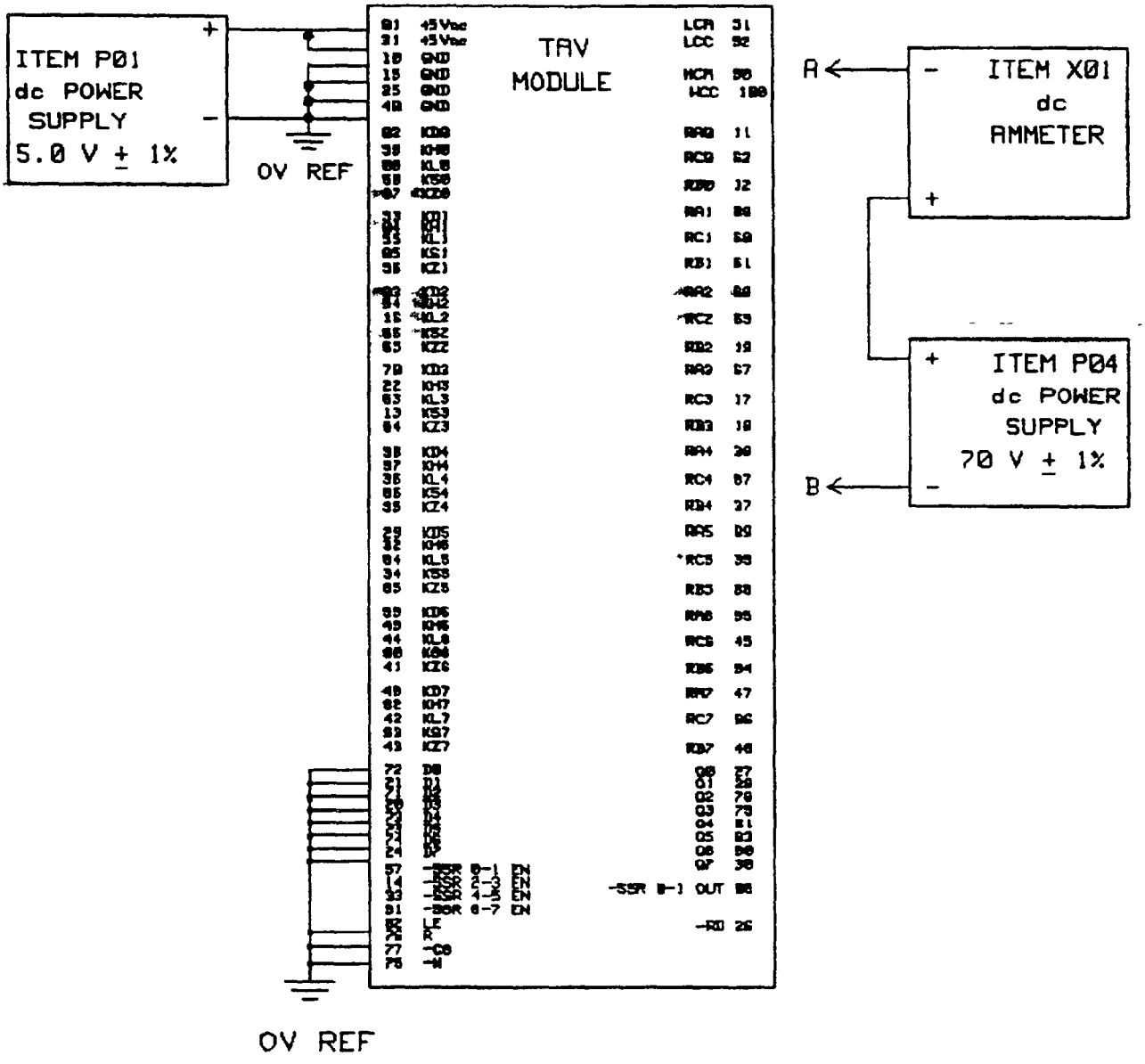


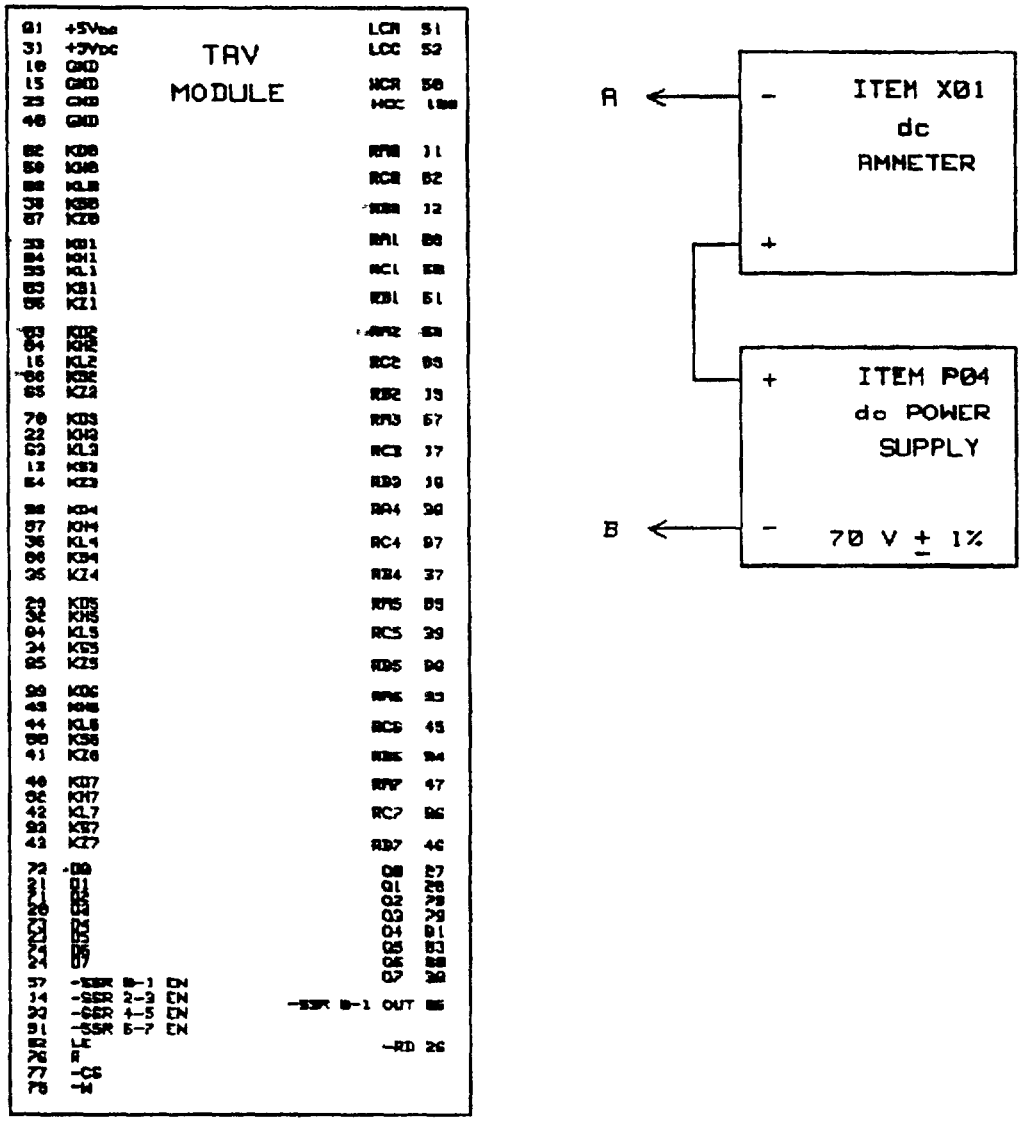
FIGURE 12. FET switch leakage current test circuit for table XXII.

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TABLE XXIII. FET switch power "OFF" leakage current test.

Test cond	Signal name	A to pin	B to pin	Forcing voltage (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Fig
W1	RA0	11	62	70	0	25	uA	13
W2	RA1	9	60					
W3	RA2	68	69					
W4	RA3	67	17					
W5	RA4	38	87					
W6	RA5	89	39					
W7	RA6	95	45					
W8	RA7	47	96					
W9	RB0	12	62					
W10	RB1	61	60					
W11	RB2	19	69					
W12	RB3	18	17					
W13	RB4	37	87					
W14	RB5	88	39					
W15	RB6	94	45					
W16	RB7	46	96	70	0	25	uA	13

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NOTE:

1. After applying the voltage, wait for the current to settle before taking the measurement.

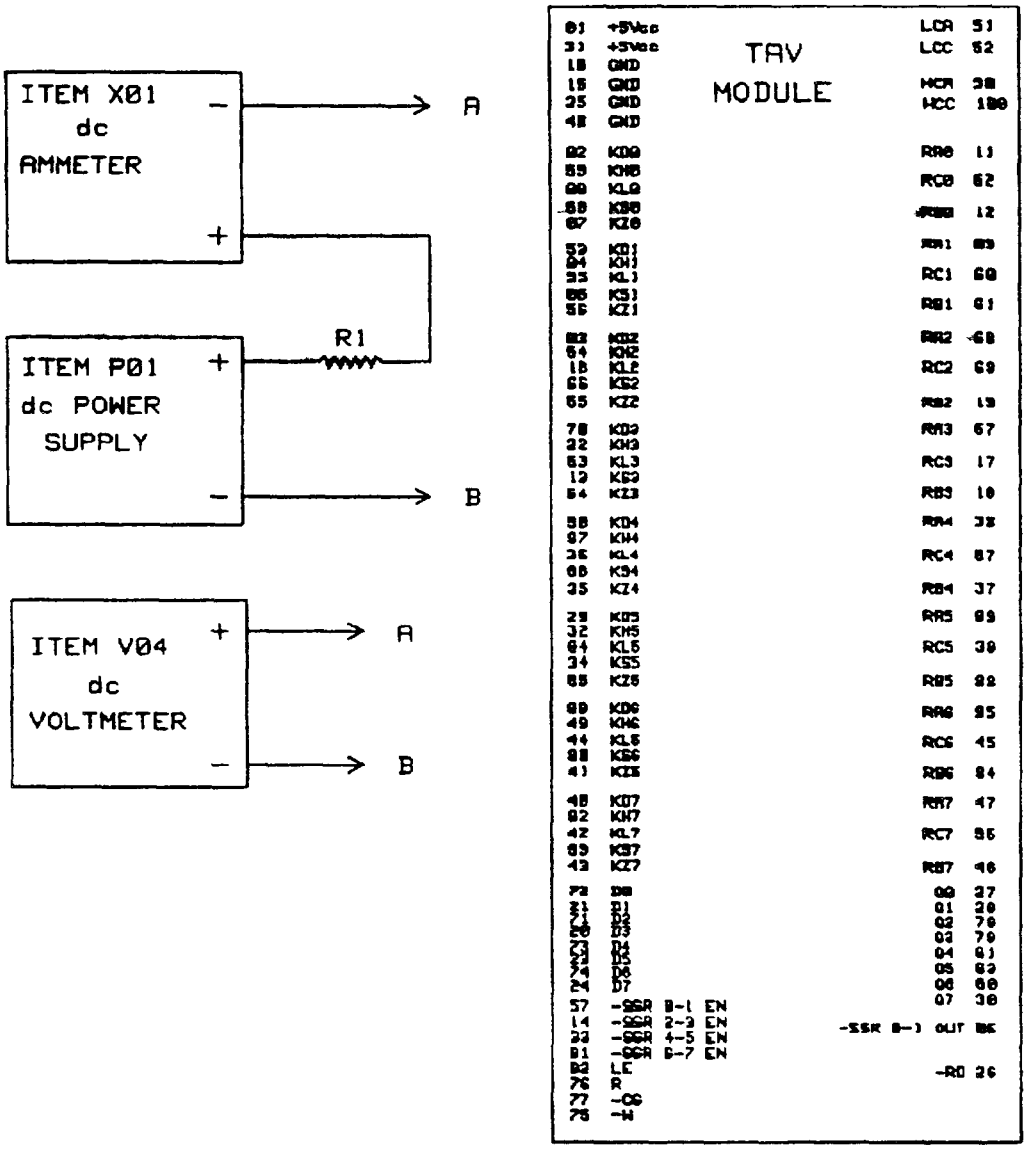
FIGURE 13. FET switch power "OFF" leakage current test circuit for table XXIII.

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TABLE XXIV. Input sense diode, forward voltage test.

Test cond	Signal name	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Fig
X1	KD0 to KH0	2	59	0.0	1.2	V	14
X2	KD1 to KH1	53	4				
X3	KD2 to KH2	3	54				
X4	KD3 to KH3	70	22				
X5	KD4 to KH4	98	97				
X6	KD5 to KH5	29	32				
X7	KD6 to KH6	99	49				
X8	KD7 to KH7	48	92	0.0	1.2	V	14

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NOTES:

1. R1 = 80 Ohm ± 1 percent, 1/4 W.

2. Apply 240 mA ± 1 percent through the diode (KDX to KHX where X = 0 to 7) as shown by adjusting the output voltage of the power supply.

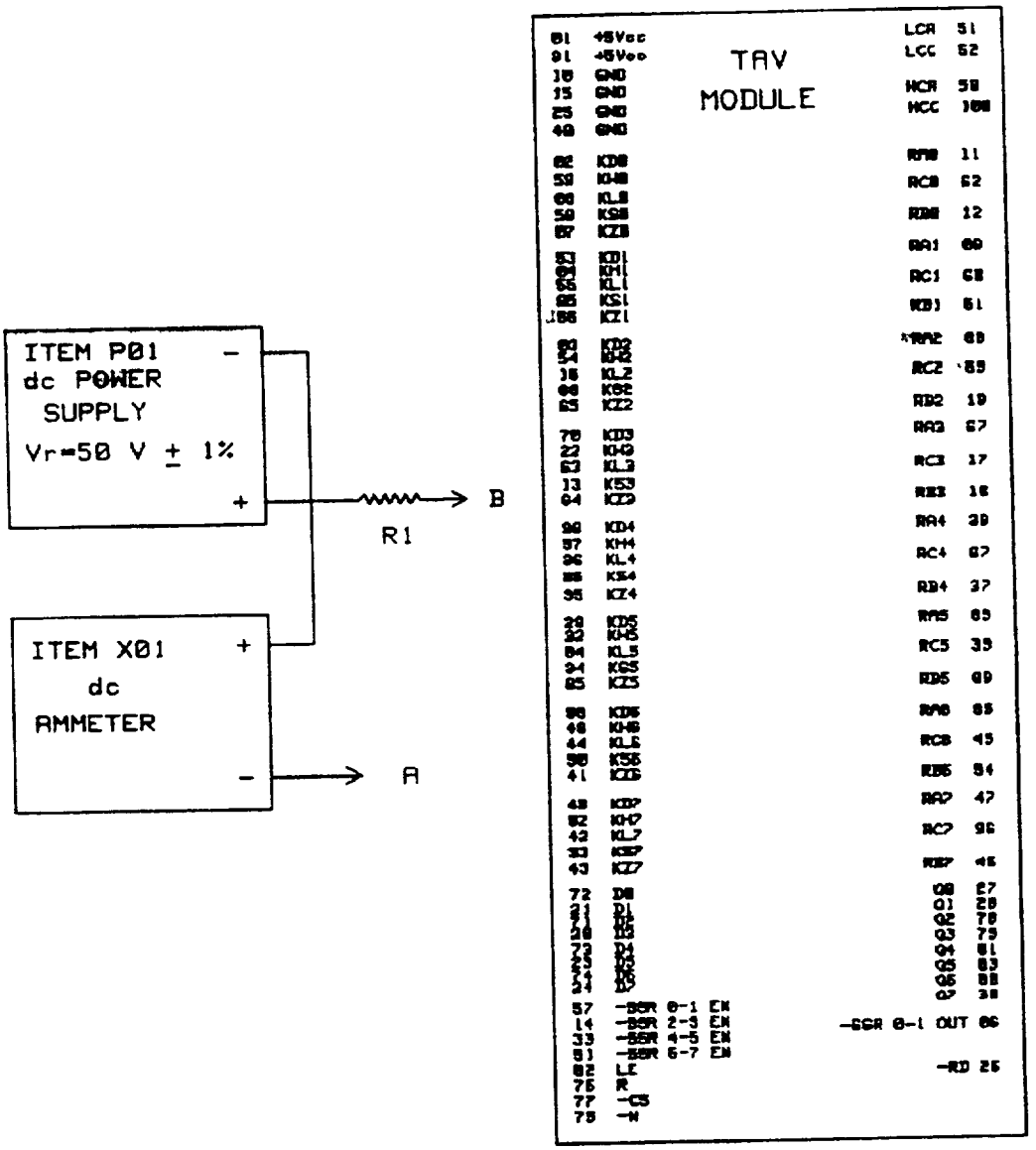
FIGURE 14. Input sense diode, forward voltage circuit for table XXIV.

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TABLE XXV. Input sense diode, reverse leakage current test.

Test cond	Signal name	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Fig
Y1	KD0 to KH0	2	59	-10	0	uA	15
Y2	KD1 to KH1	53	4				
Y3	KD2 to KH2	3	54				
Y4	KD3 to KH3	70	22				
Y5	KD4 to KH4	98	97				
Y6	KD5 to KH5	29	32				
Y7	KD6 to KH6	99	49				
Y8	KD7 to KH7	48	92	-10	0	uA	15

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NOTES:

1. R1 = 1 MOhm ± 1 percent, 1/4 W.

2. To perform this test, apply -50 V across the diode (KDX to KHX where X = 0 to 7).

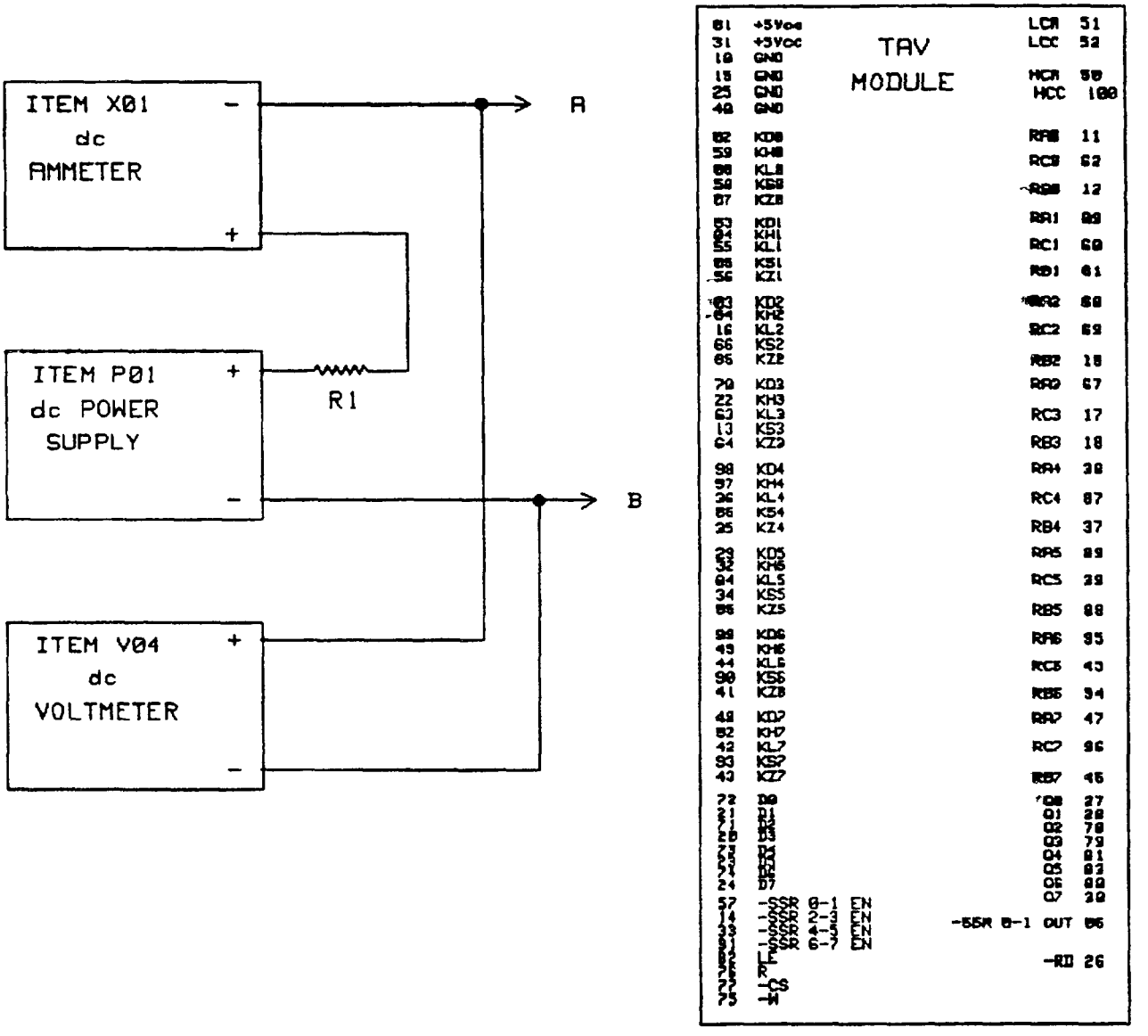
FIGURE 15. Input sense diode, reverse leakage current circuit for table XXV.

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TABLE XXVI. Zener voltage test.

Test cond	Signal name	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Fig
Z1	RA0 to RC0	11	62	73.8	90.2	V	16
Z2	RB0 to RC0	12	62				
Z3	RA1 to RC1	9	60				
Z4	RB1 to RC1	61	60				
Z5	RA2 to RC2	68	69				
Z6	RB2 to RC2	19	69				
Z7	RA3 to RC3	67	17				
Z8	RB3 to RC3	18	17				
Z9	RA4 to RC4	38	87				
Z10	RB4 to RC4	37	87				
Z11	RA5 to RC5	89	39				
Z12	RB5 to RC5	88	39				
Z13	RA6 to RC6	95	45				
Z14	RB6 to RC6	94	45				
Z15	RA7 to RC7	47	96				
Z16	RB7 to RC7	46	96	73.8	90.2	V	16

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NOTES:

1. R1 = 5 KOhm ± 1 percent, 1/4 W.

2. Apply 1.5 mA ± 1 percent through the Zener diode (RAX, RCX and RBX, RCX, where X = 0 to 7), by adjusting the voltage of Item P01.

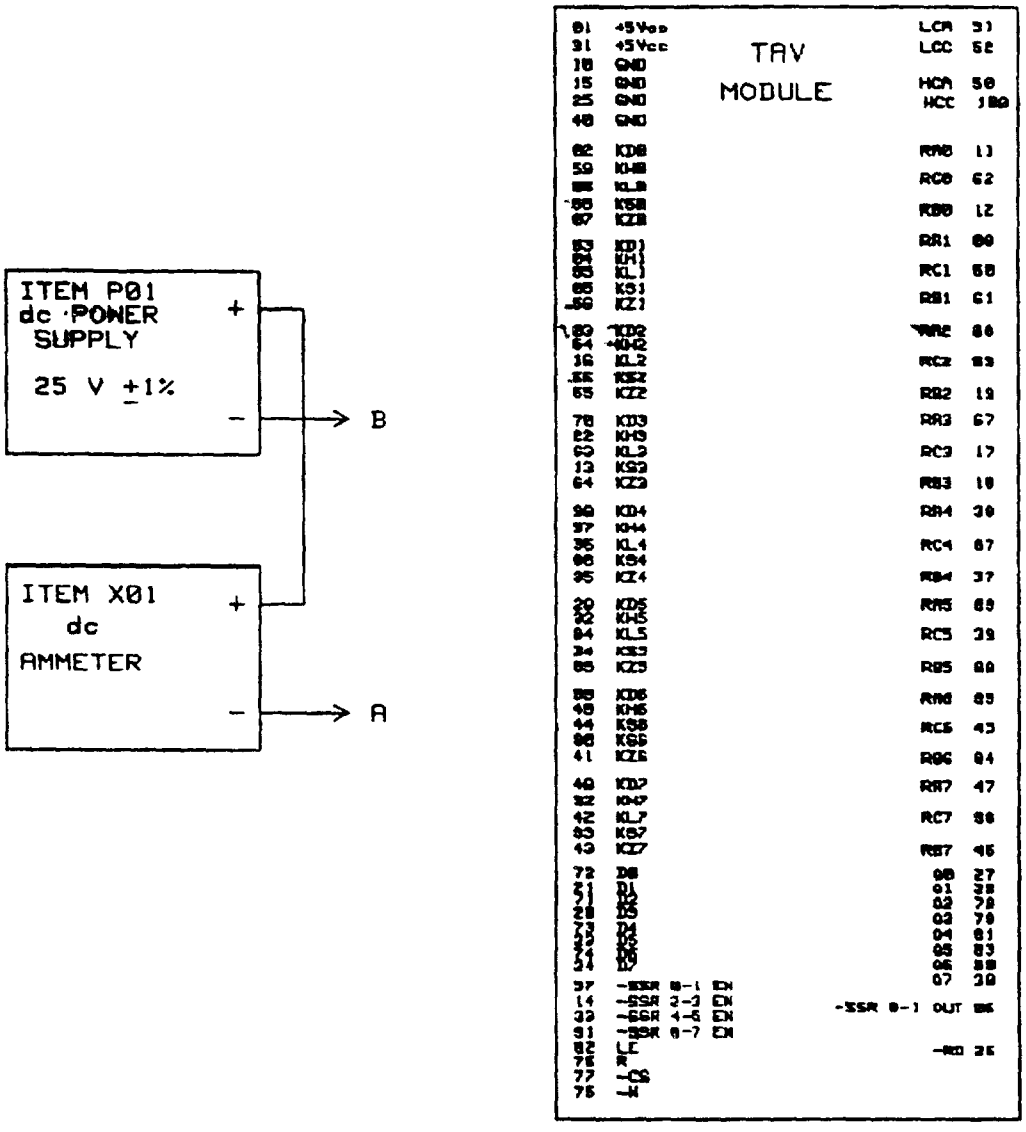
FIGURE 16. Zener voltage test circuit for table XXVI.

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TABLE XXVII. Constant current diode test.

Test cond	Signal name	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Fig
AA1	LCA to LCC	51	52	0.675	0.825	mA	17
AA2	HCA to HCC	50	100	3.51	4.60	mA	17

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NOTE:

1. Wait 15 seconds after the application of power for the diode to stabilize before making the current measurement for test condition AA2. Wait 1 second after the application of power for the diode to stabilize before making the current measurement for test condition AA1.

FIGURE 17. Constant current diode test circuit for table XXVII.

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TABLE XXVIII. Disable time (tPLZ and tPHZ) test.

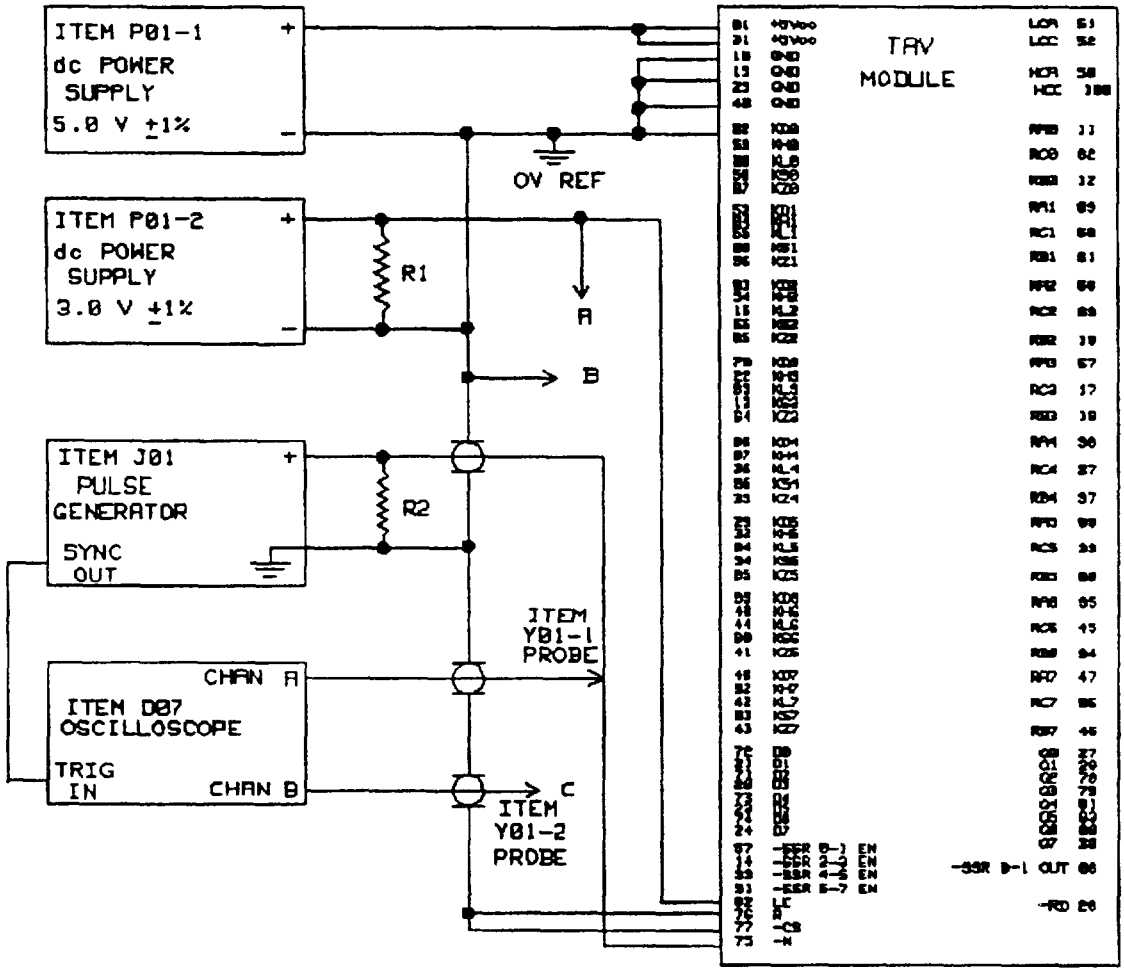
Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Figs
BB1	R to Q0,tPLZ	Open	8	27	7	69	nS	18, 19
BB2	R to Q1,tPLZ		55	28				
BB3	R to Q2,tPLZ		16	78				
BB4	R to Q3,tPLZ		63	79				
BB5	R to Q4,tPLZ		36	81				
BB6	R to Q5,tPLZ		84	83				
BB7	R to Q6,tPLZ		44	80				
BB8	R to Q7,tPLZ	Open	42	30	7	69	nS	18, 19
BB9	R to Q0,tPHZ	8	7	27	7	65	nS	18, 19
BB10	R to Q1,tPHZ	55	56	28				
BB11	R to Q2,tPHZ	16	65	78				
BB12	R to Q3,tPHZ	63	64	79				
BB13	R to Q4,tPHZ	36	35	81				
BB14	R to Q5,tPHZ	84	85	83				
BB15	R to Q6,tPHZ	44	41	80	7	65	nS	18, 19

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TABLE XXVIII. Disable time (tPLZ and tPHZ) test - Continued.

Test cond	Signal name	Connect			25 ± 5°C		Units	Figs
		A to pin	B to pin	C to pin	Initial limits			
					Min	Max		
BB16	R to Q7, tPHZ	42	43	30	7	65	nS	18, 19

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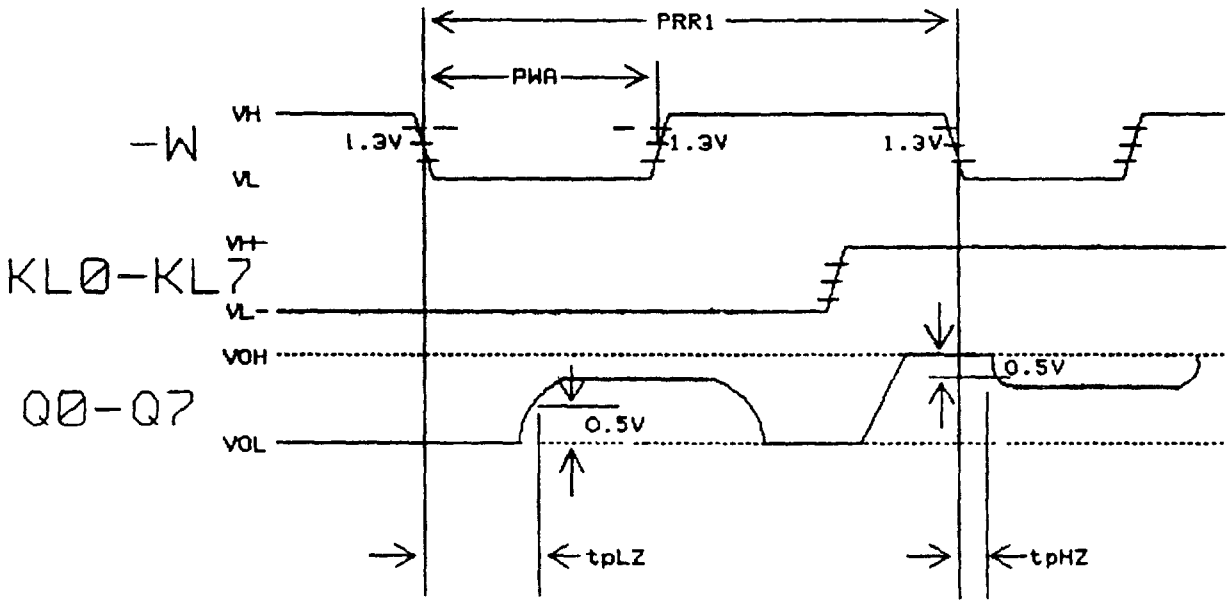
NOTES:

1. Output pins 27, 28, 78, 79, 81, 83, 80, and 30 shall be loaded as specified on figure 6.

2. R1 = R2 = 51 Ohm ± 5 percent, 1/2 W.

FIGURE 18. Disable time test circuit for table XXVIII.

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NOTE:

1. Input parameters:

   PRR1 = 1 uS minimum.
   PWA = 500 nS minimum.
   tR ≤ 15 nS, tF ≤ 15 nS (referenced 10 percent to 90 percent).

FIGURE 19. Disable time (tPLZ and tPHZ) test circuit waveforms for table XXVIII.

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TABLE XXIX. Input sense current test.

Test cond	Signal name	Connect A to pin	Connect B to pin	Item P01 voltage	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Fig
CC1	KH0	59	7	35 ± 1%	6.5	7.5	mA	20
CC2	KH1	4	56					
CC3	KH2	54	65					
CC4	KH3	22	64					
CC5	KH4	97	35					
CC6	KH5	32	85					
CC7	KH6	49	41					
CC8	KH7	92	43					
CC9	KH0	7	59					
CC10	KH1	56	4					
CC11	KH2	65	54					
CC12	KH3	64	22					
CC13	KH4	35	97					
CC14	KH5	85	32					
CC15	KH6	41	49					
CC16	KH7	43	92	35 ± 1%	6.5	7.5		
CC17	KH0	59	7	16 ± 1%	2.8	3.5		
CC18	KH1	4	56					
CC19	KH2	54	65					
CC20	KH3	22	64					
CC21	KH4	97	35	16 ± 1%	2.8	3.5	mA	20

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TABLE XXIX. Input sense current test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	Item P01 voltage	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Fig
CC22	KH5	32	85	16 ± 1%	2.8	3.5	mA	20
CC23	KH6	49	41					
CC24	KH7	92	43					
CC25	KH0	7	59					
CC26	KH1	56	4					
CC27	KH2	65	54					
CC28	KH3	64	22					
CC29	KH4	35	97					
CC30	KH5	85	32					
CC31	KH6	41	49					
CC32	KH7	43	92	16 ± 1%	2.8	3.5		
CC33	KL0	8	7	7 ± 1%	11.0	14.0		
CC34	KL1	55	56					
CC35	KL2	16	65					
CC36	KL3	63	64					
CC37	KL4	36	35					
CC38	KL5	84	85					
CC39	KL6	44	41					
CC40	KL7	42	43		11.0	14.0		
CC41	KL0	7	8		15.3	17.0		
CC42	KL1	56	55	7 ± 1%	15.3	17.0	mA	20

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TABLE XXIX. Input sense current test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	Item P01 voltage	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Fig
CC43	KL2	65	16	7 ± 1%	15.3	17.0	mA	20
CC44	KL3	64	63					
CC45	KL4	35	36					
CC46	KL5	85	84					
CC47	KL6	41	44					
CC48	KL7	43	42	7 ± 1%	15.3	17.0		
CC49	KL0	8	7	2.4 ± 1%	1.5	3.5		
CC50	KL1	55	56					
CC51	KL2	16	65					
CC52	KL3	63	64					
CC53	KL4	36	35					
CC54	KL5	84	85					
CC55	KL6	44	41					
CC56	KL7	42	43		1.5	3.5		
CC57	KL0	7	8		3.5	5.0		
CC58	KL1	56	55					
CC59	KL2	65	16					
CC60	KL3	64	63					
CC61	KL4	35	36	2.4 ± 1%	3.5	5.0	mA	20

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TABLE XXIX. Input sense current test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	Item P01 voltage	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Fig
CC62	KL5	85	84	2.4 ± 1%	3.5	5.0	mA	20
CC63	KL6	41	44					
CC64	KL7	43	42	2.4 ± 1%	3.5	5.0	mA	20

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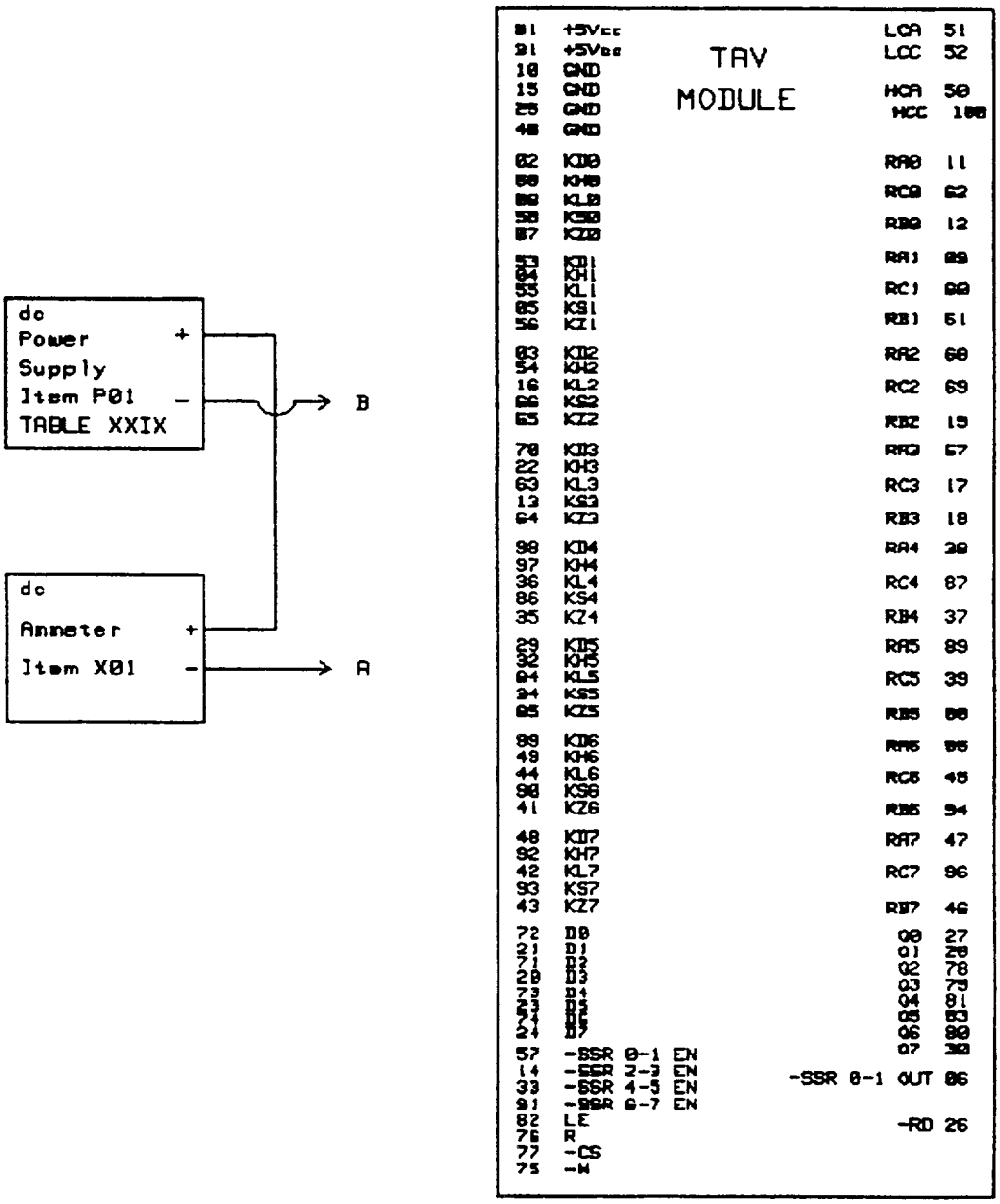


FIGURE 20. Input sense current test for table XXIX.

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	MIL-M-28787/371	

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4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

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a. NAME	b. TELEPHONE (Include Area Code)	
Technical Point of Contact (TPOC): Mr. Robert F. Dilger (Code 602A) PLEASE ADDRESS ALL CORRESPONDENCE AS FOLLOWS:	(1) Commercial TPOC: 812-854-1528	(2) AUTOVON 88-482-1528
c. ADDRESS (Include Zip Code) Commanding Officer Naval Weapons Support Center ATTN: Mr. Robert F. Dilger, Bldg. 2940W Crane, Indiana 47522-5060	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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